Air System Sizing Summary for AC-MECH/ELEC

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton) - Copy(converted)　　06/11/2026
Prepared by: Shakespeare Engineering　　11:07 AM

Air System Information

Air System Name	**AC-MECH/ELEC**		Number of zones	**1**	
Equipment Class	**SPLT AHU**		Floor Area	**330.0**	sqft
Air System Type	**SZCAV**		Location	**St George Municipal, UT, USA**	

Sizing Calculation Information

Calculation Months	**Jan to Dec**	Zone CFM Sizing	**Sum of space airflow rates**
Sizing Data	**Calculated**	Space CFM Sizing	**Individual peak space loads**

Central Cooling Coil Sizing Data

Total coil load	**0.8**	Tons	Peak coil load occurs at	**September 15:00**	
Total coil load	**9.4**	MBH	OA DB / WB	**101.0 / 63.2**	F
Sensible coil load	**9.4**	MBH	Entering DB / WB	**68.0 / 51.4**	F
Coil CFM at peak load	**655**	CFM	Leaving DB / WB	**53.5 / 45.2**	F
Sum of peak zone CFM	**655**	CFM	Resulting RH	**32**	%
Sensible heat ratio	**1.000**		Design supply temp.	**55.0**	F
CFM/Ton	**840.2**		Zone T-stat Check	**1 of 1**	OK
sqft/Ton	**423.3**		Max zone temperature deviation	**0.0**	F
BTU/(hr sqft)	**28.3**				
Water flow @ 10.0 F rise	**N/A**				

Supply Fan Sizing Data

Design CFM	**655**	CFM	Fan motor BHP	**0.36**	BHP
Design CFM/sqft	**1.99**	CFM/sqft	Fan motor kW	**0.28**	kW
			Fan total static	**2.00**	in wg

Outdoor Ventilation Air Data

Design airflow CFM	**0**	CFM	CFM/person	**0.00**	CFM/person
CFM/sqft	**0.00**	CFM/sqft			

Zone Sizing Summary for AC-MECH/ELEC
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Air System Information

Air System Name	**AC-MECH/ELEC**	Number of zones	**1**
Equipment Class	**SPLT AHU**	Floor Area	**330.0** sqft
Air System Type	**SZCAV**	Location	**St George Municipal, UT, USA**

Sizing Calculation Information

Calculation Months	**Jan to Dec**	Zone CFM Sizing	**Sum of space airflow rates**
Sizing Data	**Calculated**	Space CFM Sizing	**Individual peak space loads**

Zone Terminal Sizing Data

Zone Name	Design Supply Airflow (CFM)	Minimum Supply Airflow (CFM)	Zone CFM/sqft	Reheat Coil Capacity (MBH)	Reheat Coil Water gpm @ 20.0 F	Zone Htg Unit Coil Capacity (MBH)	Zone Htg Unit Water gpm @ 20.0 F	Mixing Box Fan Airflow (CFM)
AC-ELEC/MECH ROOM	655	655	1.99	0.0	-	0.0	-	0

Zone Peak Sensible Loads

Zone Name	Zone Cooling Sensible (MBH)	Time of Peak Sensible Cooling Load	Zone Heating Load (MBH)	Zone Floor Area (sqft)
AC-ELEC/MECH ROOM	8.3	September 15:00	0.0	330.0

Space Loads and Airflows

Zone Name / Space Name	Cooling Sensible (MBH)	Time of Peak Sensible Load	Air Flow (CFM)	Heating Load (MBH)	Floor Area (sqft)	Space CFM/sqft
AC-ELEC/MECH ROOM						
135-MECHANICAL/ELECTRICAL	8.3	September 15:00	655	0.0	330.0	1.99

1. Summary

Ventilation Sizing Method .... **Sum of Space OA Airflows**
Design Ventilation Airflow Rate .... **0** CFM

2. Space Ventilation Analysis

Zone Name / Space Name	Floor Area (sqft)	Maximum Occupants	Maximum Supply Air (CFM)	Required Outdoor Air (CFM/person)	Required Outdoor Air (CFM/sqft)	Required Outdoor Air (CFM)	Required Outdoor Air (% of supply)	Required Outdoor Air (ACH)	Uncorrected Outdoor Air (CFM)	Direct Exhaust Air (CFM)
AC-ELEC/MECH ROOM										
135-MECHANICAL/ELECTRICAL	330.0	0.0	655.2	0.00	0.00	0.0	0.0	0.00	0.0	0.0
Totals	**330.0**	**0.0**	**655.2**						**0.0**	

Table 1. System Loads

	DESIGN COOLING - SEPTEMBER 15:00			DESIGN HEATING		
	OA DB / WB 101.0 F / 63.2 F			OA DB / WB 20.2 F / 16.3 F		
COMPONENT LOADS	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**
Zone Conditioning	-	8385	0	-	971	0
Plenum Load	-	0	0	-	0	0
Return Fan Load	655 CFM	0	-	655 CFM	0	-
Ventilation Load	0 CFM	0	0	0 CFM	0	0
Supply Fan Load	655 CFM	971	-	655 CFM	-971	-
Zone Fan Coil Fans Load	-	0	-	-	0	-
>> Total System Loads	**-**	**9357**	**0**	**-**	**0**	**0**
Central Cooling Coil	-	9357	0	-	0	0
>> Total Conditioning	**-**	**9357**	**0**	**-**	**0**	**0**
Key:	**Positive values are cooling loads Negative values are heating loads**			**Positive values are heating loads Negative values are cooling loads**		

Table 2. Zone Heat Balance Loads

	DESIGN COOLING - SEPTEMBER 15:00			DESIGN HEATING		
	OA DB / WB 101.0 F / 63.2 F			OA DB / WB 20.2 F / 16.3 F		
Zone Heat Balance Component	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**
Exterior Wall Convection	158 sqft	876	-	158 sqft	71	-
Roof Convection	0 sqft	0	-	0 sqft	0	-
Window Convection	0 sqft	0	-	0 sqft	0	-
Skylight Convection	0 sqft	0	-	0 sqft	0	-
Door Convection	20 sqft	148	-	20 sqft	27	-
Floor Convection	330 sqft	489	-	330 sqft	170	-
Interior Wall Convection	616 sqft	1112	-	616 sqft	-171	-
Ceiling Convection	330 sqft	1270	-	330 sqft	-97	-
Overhead Lighting Convection	96 W	145	-	0 W	0	-
Task Lighting Convection	0 W	0	-	0 W	0	-
Electric Equipment Convection	1650 W	4223	-	0 W	0	-
People Convection	0	0	0	0	0	0
Infiltration	0 CFM	0	0	0 CFM	0	0
Miscellaneous Equipment	-	0	0	-	0	0
Air Internal Energy Change	-	0		-	0	0
Safety Factor	0% / 0%	0	0	0%	0	0
>> Total Zone Loads	**-**	**8263**	**0**	**-**	**0**	**0**
Key:	**Positive values are cooling loads Negative values are heating loads**			**Positive values are heating loads Negative values are cooling loads**		

Note 1: Surface convection line items show the combined effects of conductive heat gain to the surface and radiative heat gains absorbed at the surface which are then convected to room air.

Note 2: Lighting, equipment, and people line items include only the direct convective heat gain from the heat source to the room air. The radiative portion of the heat gain is first absorbed by surfaces in the room and then later convected from the surface to the air. Therefore the effect of the radiative portion of the heat gain is found in the surface convection line items.

Note 3: Solar heat gain is absorbed by surfaces in the room, re-radiated to other surfaces, and finally convected from the surfaces to room air. Therefore, the effect of solar heat gain is found in the surface convection line items.

Air System Sizing Summary for AC-PBX/IT

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton) - Copy(converted) 06/11/2026
Prepared by: Shakespeare Engineering 11:07 AM

Air System Information

Air System Name	**AC-PBX/IT**		Number of zones	**1**	
Equipment Class	**SPLT AHU**		Floor Area	**184.6**	sqft
Air System Type	**SZCAV**		Location	**St George Municipal, UT, USA**	

Sizing Calculation Information

Calculation Months	**Jan to Dec**	Zone CFM Sizing	**Sum of space airflow rates**
Sizing Data	**Calculated**	Space CFM Sizing	**Individual peak space loads**

Central Cooling Coil Sizing Data

Total coil load	**0.7**	Tons	Peak coil load occurs at	**July 20:00**	
Total coil load	**8.5**	MBH	OA DB / WB	**96.3 / 63.1**	F
Sensible coil load	**8.5**	MBH	Entering DB / WB	**68.0 / 53.2**	F
Coil CFM at peak load	**587**	CFM	Leaving DB / WB	**53.3 / 47.1**	F
Sum of peak zone CFM	**587**	CFM	Resulting RH	**39**	%
Sensible heat ratio	**1.000**		Design supply temp.	**55.0**	F
CFM/Ton	**828.1**		Zone T-stat Check	**1 of 1**	OK
sqft/Ton	**260.4**		Max zone temperature deviation	**0.0**	F
BTU/(hr sqft)	**46.1**				
Water flow @ 10.0 F rise	**N/A**				

Supply Fan Sizing Data

Design CFM	**587**	CFM	Fan motor BHP	**0.32**	BHP
Design CFM/sqft	**3.18**	CFM/sqft	Fan motor kW	**0.26**	kW
			Fan total static	**2.00**	in wg

Outdoor Ventilation Air Data

Design airflow CFM	**0**	CFM	CFM/person	**0.00**	CFM/person
CFM/sqft	**0.00**	CFM/sqft			

Air System Information

Air System Name	**AC-PBX/IT**	Number of zones	**1**
Equipment Class	**SPLT AHU**	Floor Area	**184.6** sqft
Air System Type	**SZCAV**	Location	**St George Municipal, UT, USA**

Sizing Calculation Information

Calculation Months	**Jan to Dec**	Zone CFM Sizing	**Sum of space airflow rates**
Sizing Data	**Calculated**	Space CFM Sizing	**Individual peak space loads**

Zone Terminal Sizing Data

Zone Name	Design Supply Airflow (CFM)	Minimum Supply Airflow (CFM)	Zone CFM/sqft	Reheat Coil Capacity (MBH)	Reheat Coil Water gpm @ 20.0 F	Zone Htg Unit Coil Capacity (MBH)	Zone Htg Unit Water gpm @ 20.0 F	Mixing Box Fan Airflow (CFM)
AC-PBX/IT ROOM	587	587	3.18	0.0	-	0.0	-	0

Zone Peak Sensible Loads

Zone Name	Zone Cooling Sensible (MBH)	Time of Peak Sensible Cooling Load	Zone Heating Load (MBH)	Zone Floor Area (sqft)
AC-PBX/IT ROOM	7.4	July 20:00	-0.1	184.6

Space Loads and Airflows

Zone Name / Space Name	Cooling Sensible (MBH)	Time of Peak Sensible Load	Air Flow (CFM)	Heating Load (MBH)	Floor Area (sqft)	Space CFM/sqft
AC-PBX/IT ROOM						
120-PBX	7.4	July 20:00	587	-0.1	184.6	3.18

Ventilation Sizing Summary for AC-PBX/IT
(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton) - Copy(converted) 06/11/2026
Prepared by: Shakespeare Engineering 11:07 AM

1. Summary

Ventilation Sizing Method .... **Sum of Space OA Airflows**
Design Ventilation Airflow Rate .... **0** CFM

2. Space Ventilation Analysis

Zone Name / Space Name	Floor Area (sqft)	Maximum Occupants	Maximum Supply Air (CFM)	Required Outdoor Air (CFM/person)	Required Outdoor Air (CFM/sqft)	Required Outdoor Air (CFM)	Required Outdoor Air (% of supply)	Required Outdoor Air (ACH)	Uncorrected Outdoor Air (CFM)	Direct Exhaust Air (CFM)
AC-PBX/IT ROOM										
120-PBX	184.6	0.0	587.1	0.00	0.00	0.0	0.0	0.00	0.0	0.0
Totals	**184.6**	**0.0**	**587.1**						**0.0**	

Table 1. System Loads

	DESIGN COOLING - JULY 20:00			DESIGN HEATING		
	OA DB / WB 96.3 F / 63.1 F			OA DB / WB 20.2 F / 16.3 F		
COMPONENT LOADS	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**
Zone Conditioning	-	7637	0	-	-163	0
Plenum Load	-	0	0	-	0	0
Return Fan Load	587 CFM	0	-	587 CFM	0	-
Ventilation Load	0 CFM	0	0	0 CFM	0	0
Supply Fan Load	587 CFM	870	-	587 CFM	-870	-
Zone Fan Coil Fans Load	-	0	-	-	0	-
>> Total System Loads	**-**	**8507**	**0**	**-**	**-1034**	**0**
Central Cooling Coil	-	8507	0	-	-1034	0
>> Total Conditioning	**-**	**8507**	**0**	**-**	**-1034**	**0**
Key:	**Positive values are cooling loads Negative values are heating loads**			**Positive values are heating loads Negative values are cooling loads**		

Table 2. Zone Heat Balance Loads

	DESIGN COOLING - JULY 20:00			DESIGN HEATING		
	OA DB / WB 96.3 F / 63.1 F			OA DB / WB 20.2 F / 16.3 F		
Zone Heat Balance Component	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**
Exterior Wall Convection	0 sqft	0	-	0 sqft	0	-
Roof Convection	0 sqft	0	-	0 sqft	0	-
Window Convection	0 sqft	0	-	0 sqft	0	-
Skylight Convection	0 sqft	0	-	0 sqft	0	-
Door Convection	0 sqft	0	-	0 sqft	0	-
Floor Convection	185 sqft	383	-	185 sqft	-26	-
Interior Wall Convection	573 sqft	1388	-	573 sqft	-107	-
Ceiling Convection	185 sqft	828	-	185 sqft	-16	-
Overhead Lighting Convection	54 W	81	-	0 W	0	-
Task Lighting Convection	0 W	0	-	0 W	0	-
Electric Equipment Convection	1846 W	4725	-	0 W	0	-
People Convection	0	0	0	0	0	0
Infiltration	0 CFM	0	0	0 CFM	0	0
Miscellaneous Equipment	-	0	0	-	0	0
Air Internal Energy Change	-	0		-	0	0
Safety Factor	0% / 0%	0	0	0%	0	0
>> Total Zone Loads	**-**	**7405**	**0**	**-**	**-149**	**0**
Key:	**Positive values are cooling loads Negative values are heating loads**			**Positive values are heating loads Negative values are cooling loads**		

Note 1: Surface convection line items show the combined effects of conductive heat gain to the surface and radiative heat gains absorbed at the surface which are then convected to room air.

Note 2: Lighting, equipment, and people line items include only the direct convective heat gain from the heat source to the room air. The radiative portion of the heat gain is first absorbed by surfaces in the room and then later convected from the surface to the air. Therefore the effect of the radiative portion of the heat gain is found in the surface convection line items.

Note 3: Solar heat gain is absorbed by surfaces in the room, re-radiated to other surfaces, and finally convected from the surfaces to room air. Therefore, the effect of solar heat gain is found in the surface convection line items.

Air System Sizing Summary for DOAS-1-CORRIDORS ONLY

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton) - Copy(converted)	06/11/2026
Prepared by: Shakespeare Engineering	11:07 AM

Air System Information

Air System Name	**DOAS-1-CORRIDORS ONLY**		Number of zones	**1**	
Equipment Class	**PKG ROOF**		Floor Area	**12123.5**	sqft
Air System Type	**MUA/DOAS**		Location	**St George Municipal, UT, USA**	

Sizing Calculation Information

Calculation Months	**Jan to Dec**	Zone CFM Sizing	**Sum of space airflow rates**
Sizing Data	**Calculated**	Space CFM Sizing	**Individual peak space loads**

Cooling Coil Sizing Data

Total coil load	**1.5**	Tons	Load occurs at	**July 15:00**	
Total coil load	**18.5**	MBH	OA DB / WB	**106.2 / 66.2**	F
Sensible coil load	**18.5**	MBH	Entering DB / WB	**100.0 / 64.4**	F
Coil CFM at July 15:00	**708**	CFM	Leaving DB / WB	**73.5 / 55.4**	F
Max coil CFM	**708**	CFM			
Sensible heat ratio	**1.000**				
Water flow @ 10.0 F rise	**N/A**				

Heating Coil Sizing Data

Max coil load	**24.4**	MBH	Load occurs at	**Design Heating**	
Coil CFM at Design Heating	**708**	CFM	Ent. DB / Lvg DB	**32.9 / 68.5**	F
Max coil CFM	**708**	CFM			
Water flow @ 20.0 F drop	**N/A**				

Ventilation Fan Sizing Data

Design CFM	**708**	CFM	Fan motor BHP	**0.39**	BHP
Design CFM/sqft	**0.06**	CFM/sqft	Fan motor kW	**0.31**	kW
			Fan total static	**2.00**	in wg

Exhaust Fan Sizing Data

Actual max CFM	**708**	CFM	Fan motor BHP	**0.00**	BHP
Standard CFM	**636**	CFM	Fan motor kW	**0.00**	kW
Actual max CFM/sqft	**0.06**	CFM/sqft	Fan total static	**0.00**	in wg

Outdoor Ventilation Air Data

Design airflow CFM	**708**	CFM	CFM/person	**143.99**	CFM/person
CFM/sqft	**0.06**	CFM/sqft			

Zone Sizing Summary for DOAS-1-CORRIDORS ONLY

(In Alternative: Default Alternative)

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Air System Information

Air System Name	DOAS-1-CORRIDORS ONLY	Number of zones	1
Equipment Class	PKG ROOF	Floor Area	12123.5 sqft
Air System Type	MUA/DOAS	Location	St George Municipal, UT, USA

Sizing Calculation Information

Calculation Months	Jan to Dec	Zone CFM Sizing	Sum of space airflow rates
Sizing Data	Calculated	Space CFM Sizing	Individual peak space loads

Zone Peak Sensible Loads

Zone Name	Zone Cooling Sensible (MBH)	Time of Peak Sensible Cooling Load	Zone Heating Load (MBH)	Zone Floor Area (sqft)
DOAS-1-CORRIDORS ONLY	58.0	July 17:00	27.3	12123.5

Space Loads and Airflows

Zone Name / Space Name	Cooling Sensible (MBH)	Time of Peak Sensible Load	Air Flow (CFM)	Heating Load (MBH)	Floor Area (sqft)	Space CFM/sqft
DOAS-1-CORRIDORS ONLY						
101-CORRIDOR (LOBBY)	1.2	June 19:00	25	0.4	413.0	0.06
108-GUEST CORRIDOR	2.1	June 10:00	12	1.7	195.2	0.06
109-GUEST CORRIDOR	2.2	July 17:00	20	0.4	336.5	0.06
113-RESTROOM	0.1	September 16:00	0	0.0	72.2	0.00
114-RESTROOM	0.2	September 16:00	0	0.0	73.1	0.00
115-SITTING AREA	2.0	July 18:00	26	0.2	185.5	0.14
118-ELEV CORRIDOR	4.0	July 17:00	33	0.7	556.4	0.06
137-GUEST CORRIDOR	2.7	June 18:00	26	1.7	433.0	0.06
138-POOL CORRIDOR	0.9	July 17:00	10	1.3	164.1	0.06
139-RESTROOM	0.1	July 20:00	0	0.0	74.1	0.00
140-RESTROOM	0.3	July 18:00	0	0.6	74.5	0.00
201-GUEST CORRIDOR	3.2	June 9:00	50	1.6	825.8	0.06
202-MECH/IDF	2.0	July 21:00	0	0.0	108.7	0.00
203-ICE	0.1	July 22:00	5	0.0	77.7	0.06
204-ELEV LOBBY	0.4	July 21:00	16	0.1	265.4	0.06
207-GUEST CORRIDOR	4.2	August 17:00	53	2.1	886.2	0.06
208-HOUSEKEEPING	1.5	August 13:00	24	0.2	239.2	0.10
301-GUEST CORRIDOR	3.2	June 10:00	51	1.5	844.0	0.06
302-MECH/IDF	2.1	July 20:00	0	0.0	112.9	0.00
303-ICE	0.1	July 22:00	5	0.0	82.7	0.06
304-ELEV LOBBY	0.4	August 16:00	16	0.0	259.7	0.06
307-GUEST CORRIDOR	4.1	August 17:00	50	1.9	832.0	0.06
308-HOUSEKEEPING	0.6	August 13:00	15	0.2	249.1	0.06
401-GUEST CORRIDOR	3.2	June 10:00	51	1.6	842.6	0.06
402-MECH/IDF	2.0	July 20:00	0	0.0	110.5	0.00
403-ICE	0.1	July 22:00	5	0.0	83.0	0.06
404-ELEV LOBBY	0.4	July 20:00	16	0.1	263.7	0.06
407-GUEST CORRIDOR	4.1	August 17:00	49	2.0	819.6	0.06
408-HOUSEKEEPING	0.6	August 13:00	15	0.2	250.9	0.06
501-GUEST CORRIDOR	3.9	July 17:00	51	3.2	847.9	0.06
502-MECH/IDF	2.4	July 17:00	0	0.2	115.0	0.00
503-ICE	0.2	July 18:00	5	0.2	81.2	0.06
504-ELEV LOBBY	0.9	July 18:00	16	0.6	259.9	0.06
507-GUEST CORRIDOR	5.9	July 17:00	50	3.6	840.9	0.06
508-HOUSEKPEEING	1.0	July 17:00	15	0.7	247.3	0.06

1. Summary

Ventilation Sizing Method .... **Sum of Space OA Airflows**
Design Ventilation Airflow Rate .... **708** CFM

2. Space Ventilation Analysis

Zone Name / Space Name	Floor Area (sqft)	Maximum Occupants	Maximum Supply Air (CFM)	Required Outdoor Air (CFM/person)	Required Outdoor Air (CFM/sqft)	Required Outdoor Air (CFM)	Required Outdoor Air (% of supply)	Required Outdoor Air (ACH)	Uncorrected Outdoor Air (CFM)	Direct Exhaust Air (CFM)
DOAS-1-CORRIDORS ONLY										
101-CORRIDOR (LOBBY)	413.0	0.0	24.8	0.00	0.06	0.0	0.0	0.00	24.8	0.0
108-GUEST CORRIDOR	195.2	0.0	11.7	0.00	0.06	0.0	0.0	0.00	11.7	0.0
109-GUEST CORRIDOR	336.5	0.0	20.2	0.00	0.06	0.0	0.0	0.00	20.2	0.0
113-RESTROOM	72.2	0.0	0.0	0.00	0.00	0.0	0.0	0.00	0.0	75.0
114-RESTROOM	73.1	0.0	0.0	0.00	0.00	0.0	0.0	0.00	0.0	75.0
115-SITTING AREA	185.5	3.0	26.1	5.00	0.06	0.0	0.0	0.00	26.1	0.0
118-ELEV CORRIDOR	556.4	0.0	33.4	0.00	0.06	0.0	0.0	0.00	33.4	0.0
137-GUEST CORRIDOR	433.0	0.0	26.0	0.00	0.06	0.0	0.0	0.00	26.0	0.0
138-POOL CORRIDOR	164.1	0.0	9.8	0.00	0.06	0.0	0.0	0.00	9.8	0.0
139-RESTROOM	74.1	0.0	0.0	0.00	0.00	0.0	0.0	0.00	0.0	75.0
140-RESTROOM	74.5	0.0	0.0	0.00	0.00	0.0	0.0	0.00	0.0	75.0
201-GUEST CORRIDOR	825.8	0.0	49.5	0.00	0.06	0.0	0.0	0.00	49.5	0.0
202-MECH/IDF	108.7	0.0	0.0	0.00	0.00	0.0	0.0	0.00	0.0	0.0
203-ICE	77.7	0.0	4.7	0.00	0.06	0.0	0.0	0.00	4.7	0.0
204-ELEV LOBBY	265.4	0.0	15.9	0.00	0.06	0.0	0.0	0.00	15.9	0.0
207-GUEST CORRIDOR	886.2	0.0	53.2	0.00	0.06	0.0	0.0	0.00	53.2	0.0
208-HOUSEKEEPING	239.2	1.9	23.9	5.00	0.06	0.0	0.0	0.00	23.9	0.0
301-GUEST CORRIDOR	844.0	0.0	50.6	0.00	0.06	0.0	0.0	0.00	50.6	0.0
302-MECH/IDF	112.9	0.0	0.0	0.00	0.00	0.0	0.0	0.00	0.0	0.0
303-ICE	82.7	0.0	5.0	0.00	0.06	0.0	0.0	0.00	5.0	0.0
304-ELEV LOBBY	259.7	0.0	15.6	0.00	0.06	0.0	0.0	0.00	15.6	0.0
307-GUEST CORRIDOR	832.0	0.0	49.9	0.00	0.06	0.0	0.0	0.00	49.9	0.0
308-HOUSEKEEPING	249.1	0.0	14.9	5.00	0.06	0.0	0.0	0.00	14.9	0.0
401-GUEST CORRIDOR	842.6	0.0	50.6	0.00	0.06	0.0	0.0	0.00	50.6	0.0
402-MECH/IDF	110.5	0.0	0.0	0.00	0.00	0.0	0.0	0.00	0.0	0.0
403-ICE	83.0	0.0	5.0	0.00	0.06	0.0	0.0	0.00	5.0	0.0
404-ELEV LOBBY	263.7	0.0	15.8	0.00	0.06	0.0	0.0	0.00	15.8	0.0
407-GUEST CORRIDOR	819.6	0.0	49.2	0.00	0.06	0.0	0.0	0.00	49.2	0.0
408-HOUSEKEEPING	250.9	0.0	15.1	5.00	0.06	0.0	0.0	0.00	15.1	0.0
501-GUEST CORRIDOR	847.9	0.0	50.9	0.00	0.06	0.0	0.0	0.00	50.9	0.0
502-MECH/IDF	115.0	0.0	0.0	0.00	0.00	0.0	0.0	0.00	0.0	0.0
503-ICE	81.2	0.0	4.9	0.00	0.06	0.0	0.0	0.00	4.9	0.0
504-ELEV LOBBY	259.9	0.0	15.6	0.00	0.06	0.0	0.0	0.00	15.6	0.0
507-GUEST CORRIDOR	840.9	0.0	50.5	0.00	0.06	0.0	0.0	0.00	50.5	0.0
508-HOUSEKPEEING	247.3	0.0	14.8	5.00	0.06	0.0	0.0	0.00	14.8	0.0

Totals	**12123.5**	**4.9**	**707.5**						**707.5**	

Air System Heat Balance Summary for DOAS-1-CORRIDORS ONLY

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton) - Copy(converted) 06/11/2026
Prepared by: Shakespeare Engineering 11:07 AM

Table 1. System Loads

	DESIGN COOLING - JULY 15:00			DESIGN HEATING		
	OA DB / WB 106.2 F / 66.2 F			OA DB / WB 20.2 F / 16.3 F		
COMPONENT LOADS	Details	Sensible [BTU/hr]	Latent [BTU/hr]	Details	Sensible [BTU/hr]	Latent [BTU/hr]
Zone Conditioning	-	6619	938	-	3579	0
Plenum Load	-	0	0	-	0	0
Exhaust Fan Load	408 CFM	0	-	408 CFM	0	-
Ventilation Load	708 CFM	10315	-935	708 CFM	21797	0
Ventilation Fan Load	708 CFM	1049	-	708 CFM	-1049	-
Zone Fan Coil Fans Load	-	0	-	-	0	-
>> Total System Loads	**-**	**17983**	**2**	**-**	**24327**	**0**
Cooling Coil	-	18458	0	-	0	0
Heating Coil	-	0	-	-	24428	-
>> Total Conditioning	**-**	**18458**	**0**	**-**	**24428**	**0**
Key:	**Positive values are cooling loads Negative values are heating loads**			**Positive values are heating loads Negative values are cooling loads**		

Table 2. Zone Heat Balance Loads

	DESIGN COOLING - JULY 15:00			DESIGN HEATING		
	OA DB / WB 106.2 F / 66.2 F			OA DB / WB 20.2 F / 16.3 F		
Zone Heat Balance Component	Details	Sensible [BTU/hr]	Latent [BTU/hr]	Details	Sensible [BTU/hr]	Latent [BTU/hr]
Exterior Wall Convection	1146 sqft	1966	-	1146 sqft	4048	-
Roof Convection	0 sqft	0	-	0 sqft	0	-
Window Convection	528 sqft	3388	-	528 sqft	10039	-
Skylight Convection	0 sqft	0	-	0 sqft	0	-
Door Convection	0 sqft	0	-	0 sqft	0	-
Floor Convection	12124 sqft	8509	-	12124 sqft	3836	-
Interior Wall Convection	31821 sqft	14580	-	31821 sqft	2206	-
Ceiling Convection	12137 sqft	13248	-	12137 sqft	7186	-
Overhead Lighting Convection	3513 W	5322	-	0 W	0	-
Task Lighting Convection	0 W	0	-	0 W	0	-
Electric Equipment Convection	2494 W	6383	-	0 W	0	-
People Convection	5	361	1007	0	0	0
Infiltration	0 CFM	0	0	0 CFM	0	0
Miscellaneous Equipment	-	0	0	-	0	0
Air Internal Energy Change	-	0		-	0	0
Safety Factor	0% / 0%	0	0	0%	0	0
>> Total Zone Loads	**-**	**53758**	**1007**	**-**	**27315**	**0**
Key:	**Positive values are cooling loads Negative values are heating loads**			**Positive values are heating loads Negative values are cooling loads**		

Note 1: Surface convection line items show the combined effects of conductive heat gain to the surface and radiative heat gains absorbed at the surface which are then convected to room air.

Note 2: Lighting, equipment, and people line items include only the direct convective heat gain from the heat source to the room air. The radiative portion of the heat gain is first absorbed by surfaces in the room and then later convected from the surface to the air. Therefore the effect of the radiative portion of the heat gain is found in the surface convection line items.

Note 3: Solar heat gain is absorbed by surfaces in the room, re-radiated to other surfaces, and finally convected from the surfaces to room air. Therefore, the effect of solar heat gain is found in the surface convection line items.

Air System Sizing Summary for HP-101-FITNESS

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton) - Copy(converted) 06/11/2026
Prepared by: Shakespeare Engineering 11:07 AM

Air System Information

Air System Name	**HP-101-FITNESS**		Number of zones	**1**	
Equipment Class	**SPLT AHU**		Floor Area	**1279.5**	sqft
Air System Type	**SZCAV**		Location	**St George Municipal, UT, USA**	

Sizing Calculation Information

Calculation Months	**Jan to Dec**	Zone CFM Sizing	**Sum of space airflow rates**
Sizing Data	**Calculated**	Space CFM Sizing	**Individual peak space loads**

Central Cooling Coil Sizing Data

Total coil load	**3.2**	Tons	Peak coil load occurs at	**July 16:00**	
Total coil load	**38.4**	MBH	OA DB / WB	**106.2 / 66.2**	F
Sensible coil load	**35.0**	MBH	Entering DB / WB	**80.5 / 65.1**	F
Coil CFM at peak load	**1464**	CFM	Leaving DB / WB	**56.4 / 56.3**	F
Sum of peak zone CFM	**1464**	CFM	Resulting RH	**66**	%
Sensible heat ratio	**0.912**		Design supply temp.	**58.0**	F
CFM/Ton	**458.0**		Zone T-stat Check	**1 of 1**	OK
sqft/Ton	**400.2**		Max zone temperature deviation	**0.0**	F
BTU/(hr sqft)	**30.0**				
Water flow @ 10.0 F rise	**N/A**				

Central Heating Coil Sizing Data

Max coil load	**21.5**	MBH	Load occurs at	**Design Heating**	
Coil CFM at Design Heating	**1464**	CFM	BTU/(hr sqft)	**16.8**	
Max coil CFM	**1464**	CFM	Ent. DB / Lvg DB	**60.1 / 75.2**	F
Water flow @ 20.0 F drop	**N/A**				

Supply Fan Sizing Data

Design CFM	**1464**	CFM	Fan motor BHP	**0.80**	BHP
Design CFM/sqft	**1.14**	CFM/sqft	Fan motor kW	**0.64**	kW
			Fan total static	**2.00**	in wg

Outdoor Ventilation Air Data

Design airflow CFM	**335**	CFM	CFM/person	**26.20**	CFM/person
CFM/sqft	**0.26**	CFM/sqft			

Air System Information

Air System Name	**HP-101-FITNESS**	Number of zones	**1**
Equipment Class	**SPLT AHU**	Floor Area	**1279.5** sqft
Air System Type	**SZCAV**	Location	**St George Municipal, UT, USA**

Sizing Calculation Information

Calculation Months	**Jan to Dec**	Zone CFM Sizing	**Sum of space airflow rates**
Sizing Data	**Calculated**	Space CFM Sizing	**Individual peak space loads**

Zone Terminal Sizing Data

Zone Name	Design Supply Airflow (CFM)	Minimum Supply Airflow (CFM)	Zone CFM/sqft	Reheat Coil Capacity (MBH)	Reheat Coil Water gpm @ 20.0 F	Zone Htg Unit Coil Capacity (MBH)	Zone Htg Unit Water gpm @ 20.0 F	Mixing Box Fan Airflow (CFM)
HP-101-FITNESS	1464	1464	1.14	0.0	-	0.0	-	0

Zone Peak Sensible Loads

Zone Name	Zone Cooling Sensible (MBH)	Time of Peak Sensible Cooling Load	Zone Heating Load (MBH)	Zone Floor Area (sqft)
HP-101-FITNESS	21.3	June 19:00	5.4	1279.5

Space Loads and Airflows

Zone Name / Space Name	Cooling Sensible (MBH)	Time of Peak Sensible Load	Air Flow (CFM)	Heating Load (MBH)	Floor Area (sqft)	Space CFM/sqft
HP-101-FITNESS						
141-FITNESS	17.4	June 19:00	1196	5.0	1117.5	1.07
142-GUEST LAUNDRY	3.9	July 18:00	268	0.4	162.0	1.66

1. Summary

Ventilation Sizing Method .... **Sum of Space OA Airflows**
Design Ventilation Airflow Rate .... **318** CFM
Design Ventilation Airflow Rate, Corrected for Exhaust Air .... **335** CFM

2. Space Ventilation Analysis

Zone Name / Space Name	Floor Area (sqft)	Maximum Occupants	Maximum Supply Air (CFM)	Required Outdoor Air (CFM/person)	Required Outdoor Air (CFM/sqft)	Required Outdoor Air (CFM)	Required Outdoor Air (% of supply)	Required Outdoor Air (ACH)	Uncorrected Outdoor Air (CFM)	Direct Exhaust Air (CFM)
HP-101-FITNESS										
141-FITNESS	1117.5	11.2	1195.8	20.00	0.06	0.0	0.0	0.00	290.6	335.3
142-GUEST LAUNDRY	162.0	1.6	268.4	5.00	0.12	0.0	0.0	0.00	27.5	0.0
Totals	**1279.5**	**12.8**	**1464.2**						**318.1**	

Air System Heat Balance Summary for HP-101-FITNESS

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton) - Copy(converted) 06/11/2026
Prepared by: Shakespeare Engineering 11:07 AM

Table 1. System Loads

	DESIGN COOLING - JULY 16:00			DESIGN HEATING		
	OA DB / WB 106.2 F / 66.2 F			OA DB / WB 20.2 F / 16.3 F		
COMPONENT LOADS	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**
Zone Conditioning	-	22026	12466	-	6740	0
Plenum Load	-	0	0	-	0	0
Return Fan Load	1129 CFM	0	-	1129 CFM	0	-
Ventilation Load	335 CFM	10720	-9252	335 CFM	16850	0
Supply Fan Load	1464 CFM	2171	-	1464 CFM	-2171	-
Zone Fan Coil Fans Load	-	0	-	-	0	-
>> Total System Loads	**-**	**34916**	**3214**	**-**	**21418**	**0**
Central Cooling Coil	-	35009	3359	-	0	0
Central Heating Coil	-	0	-	-	21490	-
>> Total Conditioning	**-**	**35009**	**3359**	**-**	**21490**	**0**
Key:	**Positive values are cooling loads Negative values are heating loads**			**Positive values are heating loads Negative values are cooling loads**		

Table 2. Zone Heat Balance Loads

	DESIGN COOLING - JULY 16:00			DESIGN HEATING		
	OA DB / WB 106.2 F / 66.2 F			OA DB / WB 20.2 F / 16.3 F		
Zone Heat Balance Component	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**
Exterior Wall Convection	198 sqft	715	-	198 sqft	429	-
Roof Convection	0 sqft	0	-	0 sqft	0	-
Window Convection	116 sqft	1012	-	116 sqft	988	-
Skylight Convection	0 sqft	0	-	0 sqft	0	-
Door Convection	0 sqft	0	-	0 sqft	0	-
Floor Convection	1280 sqft	2820	-	1280 sqft	2237	-
Interior Wall Convection	1752 sqft	3410	-	1752 sqft	677	-
Ceiling Convection	1280 sqft	4430	-	1280 sqft	1118	-
Overhead Lighting Convection	737 W	1116	-	0 W	0	-
Task Lighting Convection	0 W	0	-	0 W	0	-
Electric Equipment Convection	1928 W	4933	-	0 W	0	-
People Convection	13	2524	12918	0	0	0
Infiltration	0 CFM	0	0	0 CFM	0	0
Miscellaneous Equipment	-	0	0	-	0	0
Air Internal Energy Change	-	0		-	0	0
Safety Factor	0% / 0%	0	0	0%	0	0
>> Total Zone Loads	**-**	**20960**	**12918**	**-**	**5449**	**0**
Key:	**Positive values are cooling loads Negative values are heating loads**			**Positive values are heating loads Negative values are cooling loads**		

Note 1: Surface convection line items show the combined effects of conductive heat gain to the surface and radiative heat gains absorbed at the surface which are then convected to room air.

Note 2: Lighting, equipment, and people line items include only the direct convective heat gain from the heat source to the room air. The radiative portion of the heat gain is first absorbed by surfaces in the room and then later convected from the surface to the air. Therefore the effect of the radiative portion of the heat gain is found in the surface convection line items.

Note 3: Solar heat gain is absorbed by surfaces in the room, re-radiated to other surfaces, and finally convected from the surfaces to room air. Therefore, the effect of solar heat gain is found in the surface convection line items.

Air System Information

Air System Name	**HP-102-OFFICES**		Number of zones	**1**	
Equipment Class	**SPLT AHU**		Floor Area	**774.0**	sqft
Air System Type	**SZCAV**		Location	**St George Municipal, UT, USA**	

Sizing Calculation Information

Calculation Months	**Jan to Dec**	Zone CFM Sizing	**Sum of space airflow rates**
Sizing Data	**Calculated**	Space CFM Sizing	**Individual peak space loads**

Central Cooling Coil Sizing Data

Total coil load	**0.7**	Tons	Peak coil load occurs at	**July 16:00**	
Total coil load	**8.8**	MBH	OA DB / WB	**106.2 / 66.2**	F
Sensible coil load	**8.8**	MBH	Entering DB / WB	**80.2 / 61.7**	F
Coil CFM at peak load	**332**	CFM	Leaving DB / WB	**53.4 / 51.9**	F
Sum of peak zone CFM	**332**	CFM	Resulting RH	**50**	%
Sensible heat ratio	**1.000**		Design supply temp.	**55.0**	F
CFM/Ton	**453.1**		Zone T-stat Check	**1 of 1**	OK
sqft/Ton	**1055.0**		Max zone temperature deviation	**0.0**	F
BTU/(hr sqft)	**11.4**				
Water flow @ 10.0 F rise	**N/A**				

Central Heating Coil Sizing Data

Max coil load	**6.4**	MBH	Load occurs at	**Design Heating**	
Coil CFM at Design Heating	**332**	CFM	BTU/(hr sqft)	**8.3**	
Max coil CFM	**332**	CFM	Ent. DB / Lvg DB	**60.7 / 80.4**	F
Water flow @ 20.0 F drop	**N/A**				

Supply Fan Sizing Data

Design CFM	**332**	CFM	Fan motor BHP	**0.18**	BHP
Design CFM/sqft	**0.43**	CFM/sqft	Fan motor kW	**0.14**	kW
			Fan total static	**2.00**	in wg

Outdoor Ventilation Air Data

Design airflow CFM	**73**	CFM	CFM/person	**13.83**	CFM/person
CFM/sqft	**0.09**	CFM/sqft			

Air System Information

Air System Name	**HP-102-OFFICES**	Number of zones	**1**
Equipment Class	**SPLT AHU**	Floor Area	**774.0** sqft
Air System Type	**SZCAV**	Location	**St George Municipal, UT, USA**

Sizing Calculation Information

Calculation Months	**Jan to Dec**	Zone CFM Sizing	**Sum of space airflow rates**
Sizing Data	**Calculated**	Space CFM Sizing	**Individual peak space loads**

Zone Terminal Sizing Data

Zone Name	Design Supply Airflow (CFM)	Minimum Supply Airflow (CFM)	Zone CFM/sqft	Reheat Coil Capacity (MBH)	Reheat Coil Water gpm @ 20.0 F	Zone Htg Unit Coil Capacity (MBH)	Zone Htg Unit Water gpm @ 20.0 F	Mixing Box Fan Airflow (CFM)
HP-102-OFFICES	332	332	0.43	0.0	-	0.0	-	0

Zone Peak Sensible Loads

Zone Name	Zone Cooling Sensible (MBH)	Time of Peak Sensible Cooling Load	Zone Heating Load (MBH)	Zone Floor Area (sqft)
HP-102-OFFICES	5.3	June 20:00	2.6	774.0

Space Loads and Airflows

Zone Name / Space Name	Cooling Sensible (MBH)	Time of Peak Sensible Load	Air Flow (CFM)	Heating Load (MBH)	Floor Area (sqft)	Space CFM/sqft
HP-102-OFFICES						
119-SALES OFFICE	0.8	July 19:00	44	0.1	109.9	0.40
119-WORK AREA	2.5	July 19:00	142	0.3	407.7	0.35
122-GM OFFICE	1.0	July 18:00	58	0.7	120.3	0.48
123-CLOSET	0.1	July 20:00	5	0.0	25.2	0.20
124-RETAIL STORAGE	1.0	June 21:00	84	1.5	110.9	0.75

1. Summary

Ventilation Sizing Method .... **Sum of Space OA Airflows**
Design Ventilation Airflow Rate .... **73** CFM

2. Space Ventilation Analysis

Zone Name / Space Name	Floor Area (sqft)	Maximum Occupants	Maximum Supply Air (CFM)	Required Outdoor Air (CFM/person)	Required Outdoor Air (CFM/sqft)	Required Outdoor Air (CFM)	Required Outdoor Air (% of supply)	Required Outdoor Air (ACH)	Uncorrected Outdoor Air (CFM)	Direct Exhaust Air (CFM)
HP-102-OFFICES										
119-SALES OFFICE	109.9	1.0	43.6	5.00	0.06	0.0	0.0	0.00	11.6	0.0
119-WORK AREA	407.7	3.3	142.0	5.00	0.06	0.0	0.0	0.00	40.8	0.0
122-GM OFFICE	120.3	1.0	58.2	5.00	0.06	0.0	0.0	0.00	12.2	0.0
123-CLOSET	25.2	0.0	5.0	5.00	0.06	0.0	0.0	0.00	1.5	0.0
124-RETAIL STORAGE	110.9	0.0	83.7	5.00	0.06	0.0	0.0	0.00	6.7	0.0
Totals	**774.0**	**5.3**	**332.4**						**72.7**	

Air System Heat Balance Summary for HP-102-OFFICES

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton) - Copy(converted) 06/11/2026
Prepared by: Shakespeare Engineering 11:07 AM

Table 1. System Loads

	DESIGN COOLING - JULY 16:00			DESIGN HEATING		
	OA DB / WB 106.2 F / 66.2 F			OA DB / WB 20.2 F / 16.3 F		
COMPONENT LOADS	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**
Zone Conditioning	-	5961	1029	-	3216	0
Plenum Load	-	0	0	-	0	0
Return Fan Load	332 CFM	0	-	332 CFM	0	-
Ventilation Load	73 CFM	2325	-1025	73 CFM	3656	0
Supply Fan Load	332 CFM	493	-	332 CFM	-493	-
Zone Fan Coil Fans Load	-	0	-	-	0	-
>> Total System Loads	**-**	**8779**	**4**	**-**	**6379**	**0**
Central Cooling Coil	-	8804	0	-	0	0
Central Heating Coil	-	0	-	-	6395	-
>> Total Conditioning	**-**	**8804**	**0**	**-**	**6395**	**0**
Key:	**Positive values are cooling loads Negative values are heating loads**			**Positive values are heating loads Negative values are cooling loads**		

Table 2. Zone Heat Balance Loads

	DESIGN COOLING - JULY 16:00			DESIGN HEATING		
	OA DB / WB 106.2 F / 66.2 F			OA DB / WB 20.2 F / 16.3 F		
Zone Heat Balance Component	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**
Exterior Wall Convection	281 sqft	578	-	281 sqft	723	-
Roof Convection	0 sqft	0	-	0 sqft	0	-
Window Convection	0 sqft	0	-	0 sqft	0	-
Skylight Convection	0 sqft	0	-	0 sqft	0	-
Door Convection	0 sqft	0	-	0 sqft	0	-
Floor Convection	774 sqft	563	-	774 sqft	998	-
Interior Wall Convection	2203 sqft	1211	-	2203 sqft	441	-
Ceiling Convection	774 sqft	1151	-	774 sqft	410	-
Overhead Lighting Convection	330 W	499	-	0 W	0	-
Task Lighting Convection	0 W	0	-	0 W	0	-
Electric Equipment Convection	319 W	816	-	0 W	0	-
People Convection	5	387	1079	0	0	0
Infiltration	0 CFM	0	0	0 CFM	0	0
Miscellaneous Equipment	-	0	0	-	0	0
Air Internal Energy Change	-	0		-	0	0
Safety Factor	0% / 0%	0	0	0%	0	0
>> Total Zone Loads	**-**	**5206**	**1079**	**-**	**2573**	**0**
Key:	**Positive values are cooling loads Negative values are heating loads**			**Positive values are heating loads Negative values are cooling loads**		

Note 1: Surface convection line items show the combined effects of conductive heat gain to the surface and radiative heat gains absorbed at the surface which are then convected to room air.

Note 2: Lighting, equipment, and people line items include only the direct convective heat gain from the heat source to the room air. The radiative portion of the heat gain is first absorbed by surfaces in the room and then later convected from the surface to the air. Therefore the effect of the radiative portion of the heat gain is found in the surface convection line items.

Note 3: Solar heat gain is absorbed by surfaces in the room, re-radiated to other surfaces, and finally convected from the surfaces to room air. Therefore, the effect of solar heat gain is found in the surface convection line items.

Air System Information

Air System Name	**HP-103-BOH EMPLOYEE**		Number of zones	**1**	
Equipment Class	**SPLT AHU**		Floor Area	**593.7**	sqft
Air System Type	**SZCAV**		Location	**St George Municipal, UT, USA**	

Sizing Calculation Information

Calculation Months	**Jan to Dec**	Zone CFM Sizing	**Sum of space airflow rates**
Sizing Data	**Calculated**	Space CFM Sizing	**Individual peak space loads**

Central Cooling Coil Sizing Data

Total coil load	**1.1**	Tons	Peak coil load occurs at	**September 15:00**	
Total coil load	**13.3**	MBH	OA DB / WB	**101.0 / 63.2**	F
Sensible coil load	**13.3**	MBH	Entering DB / WB	**76.8 / 59.3**	F
Coil CFM at peak load	**584**	CFM	Leaving DB / WB	**53.7 / 50.6**	F
Sum of peak zone CFM	**584**	CFM	Resulting RH	**44**	%
Sensible heat ratio	**1.000**		Design supply temp.	**55.0**	F
CFM/Ton	**526.7**		Zone T-stat Check	**1 of 1**	OK
sqft/Ton	**535.2**		Max zone temperature deviation	**0.0**	F
BTU/(hr sqft)	**22.4**				
Water flow @ 10.0 F rise	**N/A**				

Central Heating Coil Sizing Data

Max coil load	**7.3**	MBH	Load occurs at	**Design Heating**	
Coil CFM at Design Heating	**584**	CFM	BTU/(hr sqft)	**12.3**	
Max coil CFM	**584**	CFM	Ent. DB / Lvg DB	**64.9 / 77.7**	F
Water flow @ 20.0 F drop	**N/A**				

Supply Fan Sizing Data

Design CFM	**584**	CFM	Fan motor BHP	**0.32**	BHP
Design CFM/sqft	**0.98**	CFM/sqft	Fan motor kW	**0.25**	kW
			Fan total static	**2.00**	in wg

Outdoor Ventilation Air Data

Design airflow CFM	**80**	CFM	CFM/person	**8.27**	CFM/person
CFM/sqft	**0.14**	CFM/sqft			

Zone Sizing Summary for HP-103-BOH EMPLOYEE
(In Alternative: Default Alternative)

Air System Information

Air System Name .... **HP-103-BOH EMPLOYEE**
Equipment Class .... **SPLT AHU**
Air System Type .... **SZCAV**

Number of zones **1**
Floor Area .... **593.7** sqft
Location .... **St George Municipal, UT, USA**

Sizing Calculation Information

Calculation Months .... **Jan to Dec**
Sizing Data .... **Calculated**

Zone CFM Sizing .... **Sum of space airflow rates**
Space CFM Sizing .... **Individual peak space loads**

Zone Terminal Sizing Data

Zone Name	Design Supply Airflow (CFM)	Minimum Supply Airflow (CFM)	Zone CFM/sqft	Reheat Coil Capacity (MBH)	Reheat Coil Water gpm @ 20.0 F	Zone Htg Unit Coil Capacity (MBH)	Zone Htg Unit Water gpm @ 20.0 F	Mixing Box Fan Airflow (CFM)
HP-103-BOH EMPLOYEE	584	584	0.98	0.0	-	0.0	-	0

Zone Peak Sensible Loads

Zone Name	Zone Cooling Sensible (MBH)	Time of Peak Sensible Cooling Load	Zone Heating Load (MBH)	Zone Floor Area (sqft)
HP-103-BOH EMPLOYEE	10.2	October 14:00	3.9	593.7

Space Loads and Airflows

Zone Name / Space Name	Cooling Sensible (MBH)	Time of Peak Sensible Load	Air Flow (CFM)	Heating Load (MBH)	Floor Area (sqft)	Space CFM/sqft
HP-103-BOH EMPLOYEE						
125-BOH CORRIDOR	0.3	July 19:00	18	0.1	140.3	0.13
132-BREAK ROOM	9.7	October 14:00	556	3.7	389.1	1.43
133-STAFF RR	0.2	September 17:00	10	0.0	64.3	0.15

Ventilation Sizing Summary for HP-103-BOH EMPLOYEE

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton) - Copy(converted) 06/11/2026
Prepared by: Shakespeare Engineering 11:07 AM

1. Summary

Ventilation Sizing Method .... **Sum of Space OA Airflows**
Design Ventilation Airflow Rate .... **80** CFM

2. Space Ventilation Analysis

Zone Name / Space Name	Floor Area (sqft)	Maximum Occupants	Maximum Supply Air (CFM)	Required Outdoor Air (CFM/person)	Required Outdoor Air (CFM/sqft)	Required Outdoor Air (CFM)	Required Outdoor Air (% of supply)	Required Outdoor Air (ACH)	Uncorrected Outdoor Air (CFM)	Direct Exhaust Air (CFM)
HP-103-BOH EMPLOYEE										
125-BOH CORRIDOR	140.3	0.0	18.4	0.00	0.06	0.0	0.0	0.00	8.4	0.0
132-BREAK ROOM	389.1	9.7	556.3	5.00	0.06	0.0	0.0	0.00	72.0	0.0
133-STAFF RR	64.3	0.0	9.5	0.00	0.00	0.0	0.0	0.00	0.0	75.0
Totals	**593.7**	**9.7**	**584.2**						**80.4**	

Air System Heat Balance Summary for HP-103-BOH EMPLOYEE

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton) - Copy(converted) 06/11/2026
Prepared by: Shakespeare Engineering 11:07 AM

Table 1. System Loads

	DESIGN COOLING - SEPTEMBER 15:00			DESIGN HEATING		
	OA DB / WB 101.0 F / 63.2 F			OA DB / WB 20.2 F / 16.3 F		
COMPONENT LOADS	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**
Zone Conditioning	-	10252	1114	-	4131	0
Plenum Load	-	0	0	-	0	0
Return Fan Load	509 CFM	0	-	509 CFM	0	-
Ventilation Load	80 CFM	2176	-1104	80 CFM	4041	0
Supply Fan Load	584 CFM	866	-	584 CFM	-866	-
Zone Fan Coil Fans Load	-	0	-	-	0	-
>> Total System Loads	**-**	**13294**	**10**	**-**	**7306**	**0**
Central Cooling Coil	-	13312	0	-	0	0
Central Heating Coil	-	0	-	-	7323	-
>> Total Conditioning	**-**	**13312**	**0**	**-**	**7323**	**0**
Key:	**Positive values are cooling loads Negative values are heating loads**			**Positive values are heating loads Negative values are cooling loads**		

Table 2. Zone Heat Balance Loads

	DESIGN COOLING - SEPTEMBER 15:00			DESIGN HEATING		
	OA DB / WB 101.0 F / 63.2 F			OA DB / WB 20.2 F / 16.3 F		
Zone Heat Balance Component	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**
Exterior Wall Convection	264 sqft	1433	-	264 sqft	711	-
Roof Convection	0 sqft	0	-	0 sqft	0	-
Window Convection	58 sqft	956	-	58 sqft	510	-
Skylight Convection	0 sqft	0	-	0 sqft	0	-
Door Convection	0 sqft	0	-	0 sqft	0	-
Floor Convection	594 sqft	1107	-	594 sqft	1520	-
Interior Wall Convection	1564 sqft	1868	-	1564 sqft	539	-
Ceiling Convection	594 sqft	1643	-	594 sqft	602	-
Overhead Lighting Convection	221 W	335	-	0 W	0	-
Task Lighting Convection	0 W	0	-	0 W	0	-
Electric Equipment Convection	778 W	1992	-	0 W	0	-
People Convection	10	671	1167	0	0	0
Infiltration	0 CFM	0	0	0 CFM	0	0
Miscellaneous Equipment	-	0	0	-	0	0
Air Internal Energy Change	-	0		-	0	0
Safety Factor	0% / 0%	0	0	0%	0	0
>> Total Zone Loads	**-**	**10006**	**1167**	**-**	**3882**	**0**
Key:	**Positive values are cooling loads Negative values are heating loads**			**Positive values are heating loads Negative values are cooling loads**		

Note 1: Surface convection line items show the combined effects of conductive heat gain to the surface and radiative heat gains absorbed at the surface which are then convected to room air.

Note 2: Lighting, equipment, and people line items include only the direct convective heat gain from the heat source to the room air. The radiative portion of the heat gain is first absorbed by surfaces in the room and then later convected from the surface to the air. Therefore the effect of the radiative portion of the heat gain is found in the surface convection line items.

Note 3: Solar heat gain is absorbed by surfaces in the room, re-radiated to other surfaces, and finally convected from the surfaces to room air. Therefore, the effect of solar heat gain is found in the surface convection line items.

Air System Sizing Summary for HP-104-BOH LAUNDRY

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton) - Copy(converted) 06/11/2026
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Air System Information

Air System Name	**HP-104-BOH LAUNDRY**		Number of zones	**1**	
Equipment Class	**SPLT AHU**		Floor Area	**1317.5**	sqft
Air System Type	**SZCAV**		Location	**St George Municipal, UT, USA**	

Sizing Calculation Information

Calculation Months	**Jan to Dec**	Zone CFM Sizing	**Sum of space airflow rates**
Sizing Data	**Calculated**	Space CFM Sizing	**Individual peak space loads**

Central Cooling Coil Sizing Data

Total coil load	**2.6**	Tons	Peak coil load occurs at	**August 15:00**	
Total coil load	**31.4**	MBH	OA DB / WB	**106.2 / 66.2**	F
Sensible coil load	**30.6**	MBH	Entering DB / WB	**77.8 / 62.5**	F
Coil CFM at peak load	**1293**	CFM	Leaving DB / WB	**53.9 / 53.8**	F
Sum of peak zone CFM	**1293**	CFM	Resulting RH	**55**	%
Sensible heat ratio	**0.972**		Design supply temp.	**55.0**	F
CFM/Ton	**493.5**		Zone T-stat Check	**1 of 1**	OK
sqft/Ton	**503.0**		Max zone temperature deviation	**0.0**	F
BTU/(hr sqft)	**23.9**				
Water flow @ 10.0 F rise	**N/A**				

Central Heating Coil Sizing Data

Max coil load	**14.3**	MBH	Load occurs at	**Design Heating**	
Coil CFM at Design Heating	**1293**	CFM	BTU/(hr sqft)	**10.9**	
Max coil CFM	**1293**	CFM	Ent. DB / Lvg DB	**64.5 / 75.8**	F
Water flow @ 20.0 F drop	**N/A**				

Supply Fan Sizing Data

Design CFM	**1293**	CFM	Fan motor BHP	**0.71**	BHP
Design CFM/sqft	**0.98**	CFM/sqft	Fan motor kW	**0.56**	kW
			Fan total static	**2.00**	in wg

Outdoor Ventilation Air Data

Design airflow CFM	**188**	CFM	CFM/person	**18.99**	CFM/person
CFM/sqft	**0.14**	CFM/sqft			

Air System Information

Air System Name .... **HP-104-BOH LAUNDRY**
Equipment Class .... **SPLT AHU**
Air System Type .... **SZCAV**

Number of zones **1**
Floor Area .... **1317.5** sqft
Location .... **St George Municipal, UT, USA**

Sizing Calculation Information

Calculation Months .... **Jan to Dec**
Sizing Data .... **Calculated**

Zone CFM Sizing .... **Sum of space airflow rates**
Space CFM Sizing .... **Individual peak space loads**

Zone Terminal Sizing Data

Zone Name	Design Supply Airflow (CFM)	Minimum Supply Airflow (CFM)	Zone CFM/sqft	Reheat Coil Capacity (MBH)	Reheat Coil Water gpm @ 20.0 F	Zone Htg Unit Coil Capacity (MBH)	Zone Htg Unit Water gpm @ 20.0 F	Mixing Box Fan Airflow (CFM)
HP-104-BOH LAUNDRY	1293	1293	0.98	0.0	-	0.0	-	0

Zone Peak Sensible Loads

Zone Name	Zone Cooling Sensible (MBH)	Time of Peak Sensible Cooling Load	Zone Heating Load (MBH)	Zone Floor Area (sqft)
HP-104-BOH LAUNDRY	21.9	August 14:00	6.0	1317.5

Space Loads and Airflows

Zone Name / Space Name	Cooling Sensible (MBH)	Time of Peak Sensible Load	Air Flow (CFM)	Heating Load (MBH)	Floor Area (sqft)	Space CFM/sqft
HP-104-BOH LAUNDRY						
126-BOH LAUNDRY	14.5	August 13:00	831	2.7	871.6	0.95
127 - DRYER ENCLOSURE	5.6	September 16:00	318	1.1	117.8	2.70
128-LINEN STORAGE	1.8	August 13:00	119	2.1	171.4	0.69
129-STORAGE	0.2	July 20:00	9	0.1	48.6	0.18
130-LINEN CLOSET	0.3	July 20:00	17	0.1	108.1	0.15

Ventilation Sizing Summary for HP-104-BOH LAUNDRY

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton) - Copy(converted) 06/11/2026
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1. Summary

Ventilation Sizing Method .... **Sum of Space OA Airflows**
Design Ventilation Airflow Rate .... **188** CFM

2. Space Ventilation Analysis

Zone Name / Space Name	Floor Area (sqft)	Maximum Occupants	Maximum Supply Air (CFM)	Required Outdoor Air (CFM/person)	Required Outdoor Air (CFM/sqft)	Required Outdoor Air (CFM)	Required Outdoor Air (% of supply)	Required Outdoor Air (ACH)	Uncorrected Outdoor Air (CFM)	Direct Exhaust Air (CFM)
HP-104-BOH LAUNDRY										
126-BOH LAUNDRY	871.6	8.7	830.6	5.00	0.12	0.0	0.0	0.00	148.2	0.0
127 - DRYER ENCLOSURE	117.8	1.2	318.0	5.00	0.12	0.0	0.0	0.00	20.0	0.0
128-LINEN STORAGE	171.4	0.0	118.7	5.00	0.06	0.0	0.0	0.00	10.3	0.0
129-STORAGE	48.6	0.0	8.7	5.00	0.06	0.0	0.0	0.00	2.9	0.0
130-LINEN CLOSET	108.1	0.0	16.5	5.00	0.06	0.0	0.0	0.00	6.5	0.0
Totals	**1317.5**	**9.9**	**1292.5**						**187.9**	

Air System Heat Balance Summary for HP-104-BOH LAUNDRY

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton) - Copy(converted)
Prepared by: Shakespeare Engineering

Table 1. System Loads

	DESIGN COOLING - AUGUST 15:00			DESIGN HEATING		
	OA DB / WB 106.2 F / 66.2 F			OA DB / WB 20.2 F / 16.3 F		
COMPONENT LOADS	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**
Zone Conditioning	-	22543	4314	-	6732	0
Plenum Load	-	0	0	-	0	0
Return Fan Load	1293 CFM	0	-	1293 CFM	0	-
Ventilation Load	188 CFM	6033	-3441	188 CFM	9443	0
Supply Fan Load	1293 CFM	1916	-	1293 CFM	-1916	-
Zone Fan Coil Fans Load	-	0	-	-	0	-
>> Total System Loads	**-**	**30492**	**873**	**-**	**14259**	**0**
Central Cooling Coil	-	30551	880	-	0	0
Central Heating Coil	-	0	-	-	14299	-
>> Total Conditioning	**-**	**30551**	**880**	**-**	**14299**	**0**
Key:	**Positive values are cooling loads Negative values are heating loads**			**Positive values are heating loads Negative values are cooling loads**		

Table 2. Zone Heat Balance Loads

	DESIGN COOLING - AUGUST 15:00			DESIGN HEATING		
	OA DB / WB 106.2 F / 66.2 F			OA DB / WB 20.2 F / 16.3 F		
Zone Heat Balance Component	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**
Exterior Wall Convection	680 sqft	2658	-	680 sqft	1630	-
Roof Convection	0 sqft	0	-	0 sqft	0	-
Window Convection	0 sqft	0	-	0 sqft	0	-
Skylight Convection	0 sqft	0	-	0 sqft	0	-
Door Convection	0 sqft	0	-	0 sqft	0	-
Floor Convection	1318 sqft	1963	-	1318 sqft	2883	-
Interior Wall Convection	2551 sqft	2813	-	2551 sqft	596	-
Ceiling Convection	1318 sqft	3115	-	1318 sqft	933	-
Overhead Lighting Convection	467 W	707	-	0 W	0	-
Task Lighting Convection	0 W	0	-	0 W	0	-
Electric Equipment Convection	3793 W	9706	-	0 W	0	-
People Convection	10	876	4502	0	0	0
Infiltration	0 CFM	0	0	0 CFM	0	0
Miscellaneous Equipment	-	0	0	-	0	0
Air Internal Energy Change	-	0		-	0	0
Safety Factor	0% / 0%	0	0	0%	0	0
>> Total Zone Loads	**-**	**21837**	**4502**	**-**	**6042**	**0**
Key:	**Positive values are cooling loads Negative values are heating loads**			**Positive values are heating loads Negative values are cooling loads**		

Note 1: Surface convection line items show the combined effects of conductive heat gain to the surface and radiative heat gains absorbed at the surface which are then convected to room air.

Note 2: Lighting, equipment, and people line items include only the direct convective heat gain from the heat source to the room air. The radiative portion of the heat gain is first absorbed by surfaces in the room and then later convected from the surface to the air. Therefore the effect of the radiative portion of the heat gain is found in the surface convection line items.

Note 3: Solar heat gain is absorbed by surfaces in the room, re-radiated to other surfaces, and finally convected from the surfaces to room air. Therefore, the effect of solar heat gain is found in the surface convection line items.

Air System Sizing Summary for HP-105-LOBBY/MARKET
(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton) - Copy(converted) 06/11/2026
Prepared by: Shakespeare Engineering 11:07 AM

Air System Information

Air System Name	**HP-105-LOBBY/MARKET**		Number of zones	**1**	
Equipment Class	**SPLT AHU**		Floor Area	**1301.9**	sqft
Air System Type	**SZCAV**		Location	**St George Municipal, UT, USA**	

Sizing Calculation Information

Calculation Months	**Jan to Dec**	Zone CFM Sizing	**Sum of space airflow rates**
Sizing Data	**Calculated**	Space CFM Sizing	**Individual peak space loads**

Central Cooling Coil Sizing Data

Total coil load	**2.5**	Tons	Peak coil load occurs at	**July 16:00**	
Total coil load	**29.6**	MBH	OA DB / WB	**106.2 / 66.2**	F
Sensible coil load	**29.0**	MBH	Entering DB / WB	**83.6 / 63.3**	F
Coil CFM at peak load	**917**	CFM	Leaving DB / WB	**51.7 / 51.6**	F
Sum of peak zone CFM	**917**	CFM	Resulting RH	**55**	%
Sensible heat ratio	**0.980**		Design supply temp.	**55.0**	F
CFM/Ton	**371.9**		Zone T-stat Check	**1 of 1**	OK
sqft/Ton	**527.8**		Max zone temperature deviation	**0.0**	F
BTU/(hr sqft)	**22.7**				
Water flow @ 10.0 F rise	**N/A**				

Central Heating Coil Sizing Data

Max coil load	**23.4**	MBH	Load occurs at	**Design Heating**	
Coil CFM at Design Heating	**917**	CFM	BTU/(hr sqft)	**18.0**	
Max coil CFM	**917**	CFM	Ent. DB / Lvg DB	**55.2 / 81.3**	F
Water flow @ 20.0 F drop	**N/A**				

Supply Fan Sizing Data

Design CFM	**917**	CFM	Fan motor BHP	**0.50**	BHP
Design CFM/sqft	**0.70**	CFM/sqft	Fan motor kW	**0.40**	kW
			Fan total static	**2.00**	in wg

Outdoor Ventilation Air Data

Design airflow CFM	**298**	CFM	CFM/person	**10.17**	CFM/person
CFM/sqft	**0.23**	CFM/sqft			

Air System Information

Air System Name: **HP-105-LOBBY/MARKET**
Equipment Class: **SPLT AHU**
Air System Type: **SZCAV**

Number of zones: **1**
Floor Area: **1301.9** sqft
Location: **St George Municipal, UT, USA**

Sizing Calculation Information

Calculation Months: **Jan to Dec**
Sizing Data: **Calculated**

Zone CFM Sizing: **Sum of space airflow rates**
Space CFM Sizing: **Individual peak space loads**

Zone Terminal Sizing Data

Zone Name	Design Supply Airflow (CFM)	Minimum Supply Airflow (CFM)	Zone CFM/sqft	Reheat Coil Capacity (MBH)	Reheat Coil Water gpm @ 20.0 F	Zone Htg Unit Coil Capacity (MBH)	Zone Htg Unit Water gpm @ 20.0 F	Mixing Box Fan Airflow (CFM)
HP-105-LOBBY/MARKET	917	917	0.70	0.0	-	0.0	-	0

Zone Peak Sensible Loads

Zone Name	Zone Cooling Sensible (MBH)	Time of Peak Sensible Cooling Load	Zone Heating Load (MBH)	Zone Floor Area (sqft)
HP-105-LOBBY/MARKET	16.0	July 18:00	8.0	1301.9

Space Loads and Airflows

Zone Name / Space Name	Cooling Sensible (MBH)	Time of Peak Sensible Load	Air Flow (CFM)	Heating Load (MBH)	Floor Area (sqft)	Space CFM/sqft
HP-105-LOBBY/MARKET						
101/102-MARKET/RECEPTION	10.1	June 18:00	579	5.2	651.0	0.89
103-LOBBY (entry)	5.9	July 17:00	338	2.8	650.9	0.52

Ventilation Sizing Summary for HP-105-LOBBY/MARKET
(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton) - Copy(converted) 06/11/2026
Prepared by: Shakespeare Engineering 11:07 AM

1. Summary

Ventilation Sizing Method .... **Sum of Space OA Airflows**
Design Ventilation Airflow Rate .... **298** CFM

2. Space Ventilation Analysis

Zone Name / Space Name	Floor Area (sqft)	Maximum Occupants	Maximum Supply Air (CFM)	Required Outdoor Air (CFM/person)	Required Outdoor Air (CFM/sqft)	Required Outdoor Air (CFM)	Required Outdoor Air (% of supply)	Required Outdoor Air (ACH)	Uncorrected Outdoor Air (CFM)	Direct Exhaust Air (CFM)
HP-105-LOBBY/MARKET										
101/102-MARKET/RECEPTION	651.0	19.5	578.9	7.50	0.06	0.0	0.0	0.00	185.7	0.0
103-LOBBY (entry)	650.9	9.8	338.5	7.50	0.06	0.0	0.0	0.00	112.2	0.0
Totals	**1301.9**	**29.3**	**917.4**						**297.9**	

Air System Heat Balance Summary for HP-105-LOBBY/MARKET

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton) - Copy(converted) 06/11/2026
Prepared by: Shakespeare Engineering 11:07 AM

Table 1. System Loads

	DESIGN COOLING - JULY 16:00			DESIGN HEATING		
	OA DB / WB 106.2 F / 66.2 F			OA DB / WB 20.2 F / 16.3 F		
COMPONENT LOADS	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**
Zone Conditioning	-	18042	5939	-	9699	0
Plenum Load	-	0	0	-	0	0
Return Fan Load	917 CFM	0	-	917 CFM	0	-
Ventilation Load	298 CFM	9525	-5486	298 CFM	14973	0
Supply Fan Load	917 CFM	1360	-	917 CFM	-1360	-
Zone Fan Coil Fans Load	-	0	-	-	0	-
>> Total System Loads	**-**	**28927**	**453**	**-**	**23311**	**0**
Central Cooling Coil	-	29014	585	-	0	0
Central Heating Coil	-	0	-	-	23375	-
>> Total Conditioning	**-**	**29014**	**585**	**-**	**23375**	**0**
Key:	**Positive values are cooling loads Negative values are heating loads**			**Positive values are heating loads Negative values are cooling loads**		

Table 2. Zone Heat Balance Loads

	DESIGN COOLING - JULY 16:00			DESIGN HEATING		
	OA DB / WB 106.2 F / 66.2 F			OA DB / WB 20.2 F / 16.3 F		
Zone Heat Balance Component	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**
Exterior Wall Convection	508 sqft	1939	-	508 sqft	1346	-
Roof Convection	0 sqft	0	-	0 sqft	0	-
Window Convection	116 sqft	1396	-	116 sqft	1001	-
Skylight Convection	0 sqft	0	-	0 sqft	0	-
Door Convection	0 sqft	0	-	0 sqft	0	-
Floor Convection	1302 sqft	3254	-	1302 sqft	3705	-
Interior Wall Convection	497 sqft	915	-	497 sqft	300	-
Ceiling Convection	1302 sqft	4481	-	1302 sqft	1655	-
Overhead Lighting Convection	448 W	679	-	0 W	0	-
Task Lighting Convection	0 W	0	-	0 W	0	-
Electric Equipment Convection	325 W	833	-	0 W	0	-
People Convection	29	2154	6008	0	0	0
Infiltration	0 CFM	0	0	0 CFM	0	0
Miscellaneous Equipment	-	0	0	-	0	0
Air Internal Energy Change	-	0		-	0	0
Safety Factor	0% / 0%	0	0	0%	0	0
>> Total Zone Loads	**-**	**15650**	**6008**	**-**	**8007**	**0**
Key:	**Positive values are cooling loads Negative values are heating loads**			**Positive values are heating loads Negative values are cooling loads**		

Note 1: Surface convection line items show the combined effects of conductive heat gain to the surface and radiative heat gains absorbed at the surface which are then convected to room air.

Note 2: Lighting, equipment, and people line items include only the direct convective heat gain from the heat source to the room air. The radiative portion of the heat gain is first absorbed by surfaces in the room and then later convected from the surface to the air. Therefore the effect of the radiative portion of the heat gain is found in the surface convection line items.

Note 3: Solar heat gain is absorbed by surfaces in the room, re-radiated to other surfaces, and finally convected from the surfaces to room air. Therefore, the effect of solar heat gain is found in the surface convection line items.

Air System Sizing Summary for HP-106-DINING LARGE

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton) - Copy(converted) 06/11/2026
Prepared by: Shakespeare Engineering 11:07 AM

Air System Information

Air System Name	**HP-106-DINING LARGE**		Number of zones	**1**	
Equipment Class	**SPLT AHU**		Floor Area	**1092.2**	sqft
Air System Type	**SZCAV**		Location	**St George Municipal, UT, USA**	

Sizing Calculation Information

Calculation Months	**Jan to Dec**	Zone CFM Sizing	**Sum of space airflow rates**
Sizing Data	**Calculated**	Space CFM Sizing	**Individual peak space loads**

Central Cooling Coil Sizing Data

Total coil load	**2.9**	Tons	Peak coil load occurs at	**July 16:00**	
Total coil load	**34.6**	MBH	OA DB / WB	**106.2 / 66.2**	F
Sensible coil load	**34.6**	MBH	Entering DB / WB	**93.7 / 63.6**	F
Coil CFM at peak load	**832**	CFM	Leaving DB / WB	**51.6 / 48.0**	F
Sum of peak zone CFM	**832**	CFM	Resulting RH	**44**	%
Sensible heat ratio	**1.000**		Design supply temp.	**55.0**	F
CFM/Ton	**289.0**		Zone T-stat Check	**1 of 1**	OK
sqft/Ton	**379.2**		Max zone temperature deviation	**0.0**	F
BTU/(hr sqft)	**31.6**				
Water flow @ 10.0 F rise	**N/A**				

Central Heating Coil Sizing Data

Max coil load	**34.5**	MBH	Load occurs at	**Design Heating**	
Coil CFM at Design Heating	**832**	CFM	BTU/(hr sqft)	**31.6**	
Max coil CFM	**832**	CFM	Ent. DB / Lvg DB	**39.4 / 81.9**	F
Water flow @ 20.0 F drop	**N/A**				

Supply Fan Sizing Data

Design CFM	**832**	CFM	Fan motor BHP	**0.46**	BHP
Design CFM/sqft	**0.76**	CFM/sqft	Fan motor kW	**0.36**	kW
			Fan total static	**2.00**	in wg

Outdoor Ventilation Air Data

Design airflow CFM	**524**	CFM	CFM/person	**12.00**	CFM/person
CFM/sqft	**0.48**	CFM/sqft			

Air System Information

Air System Name	**HP-106-DINING LARGE**	Number of zones	**1**
Equipment Class	**SPLT AHU**	Floor Area	**1092.2** sqft
Air System Type	**SZCAV**	Location	**St George Municipal, UT, USA**

Sizing Calculation Information

Calculation Months	**Jan to Dec**	Zone CFM Sizing	**Sum of space airflow rates**
Sizing Data	**Calculated**	Space CFM Sizing	**Individual peak space loads**

Zone Terminal Sizing Data

Zone Name	Design Supply Airflow (CFM)	Minimum Supply Airflow (CFM)	Zone CFM/sqft	Reheat Coil Capacity (MBH)	Reheat Coil Water gpm @ 20.0 F	Zone Htg Unit Coil Capacity (MBH)	Zone Htg Unit Water gpm @ 20.0 F	Mixing Box Fan Airflow (CFM)
HP-106-DINING LARGE	832	832	0.76	0.0	-	0.0	-	0

Zone Peak Sensible Loads

Zone Name	Zone Cooling Sensible (MBH)	Time of Peak Sensible Cooling Load	Zone Heating Load (MBH)	Zone Floor Area (sqft)
HP-106-DINING LARGE	14.5	July 16:00	7.9	1092.2

Space Loads and Airflows

Zone Name / Space Name	Cooling Sensible (MBH)	Time of Peak Sensible Load	Air Flow (CFM)	Heating Load (MBH)	Floor Area (sqft)	Space CFM/sqft
HP-106-DINING LARGE						
105A DINING	14.5	July 16:00	832	7.9	1092.2	0.76

Ventilation Sizing Summary for HP-106-DINING LARGE

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton) - Copy(converted)	06/11/2026
Prepared by: Shakespeare Engineering	11:07 AM

1. Summary

Ventilation Sizing Method .... **Sum of Space OA Airflows**

Design Ventilation Airflow Rate .... **524** CFM

2. Space Ventilation Analysis

Zone Name / Space Name	Floor Area (sqft)	Maximum Occupants	Maximum Supply Air (CFM)	Required Outdoor Air (CFM/person)	Required Outdoor Air (CFM/sqft)	Required Outdoor Air (CFM)	Required Outdoor Air (% of supply)	Required Outdoor Air (ACH)	Uncorrected Outdoor Air (CFM)	Direct Exhaust Air (CFM)
HP-106-DINING LARGE										
105A DINING	1092.2	43.7	832.4	7.50	0.18	0.0	0.0	0.00	524.3	0.0
Totals	**1092.2**	**43.7**	**832.4**						**524.3**	

Air System Heat Balance Summary for HP-106-DINING LARGE

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton) - Copy(converted) 06/11/2026
Prepared by: Shakespeare Engineering 11:07 AM

Table 1. System Loads

	DESIGN COOLING - JULY 16:00			DESIGN HEATING		
	OA DB / WB 106.2 F / 66.2 F			OA DB / WB 20.2 F / 16.3 F		
COMPONENT LOADS	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**
Zone Conditioning	-	16375	5046	-	9240	0
Plenum Load	-	0	0	-	0	0
Return Fan Load	832 CFM	0	-	832 CFM	0	-
Ventilation Load	524 CFM	16762	-5046	524 CFM	26348	0
Supply Fan Load	832 CFM	1234	-	832 CFM	-1234	-
Zone Fan Coil Fans Load	-	0	-	-	0	-
>> Total System Loads	**-**	**34371**	**0**	**-**	**34354**	**0**
Central Cooling Coil	-	34562	0	-	0	0
Central Heating Coil	-	0	-	-	34466	-
>> Total Conditioning	**-**	**34562**	**0**	**-**	**34466**	**0**
Key:	**Positive values are cooling loads Negative values are heating loads**			**Positive values are heating loads Negative values are cooling loads**		

Table 2. Zone Heat Balance Loads

	DESIGN COOLING - JULY 16:00			DESIGN HEATING		
	OA DB / WB 106.2 F / 66.2 F			OA DB / WB 20.2 F / 16.3 F		
Zone Heat Balance Component	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**
Exterior Wall Convection	140 sqft	498	-	140 sqft	369	-
Roof Convection	0 sqft	0	-	0 sqft	0	-
Window Convection	329 sqft	2649	-	329 sqft	2882	-
Skylight Convection	0 sqft	0	-	0 sqft	0	-
Door Convection	0 sqft	0	-	0 sqft	0	-
Floor Convection	1092 sqft	2653	-	1092 sqft	3133	-
Interior Wall Convection	0 sqft	0	-	0 sqft	0	-
Ceiling Convection	1092 sqft	3845	-	1092 sqft	1535	-
Overhead Lighting Convection	317 W	480	-	0 W	0	-
Task Lighting Convection	0 W	0	-	0 W	0	-
Electric Equipment Convection	546 W	1398	-	0 W	0	-
People Convection	44	3015	5243	0	0	0
Infiltration	0 CFM	0	0	0 CFM	0	0
Miscellaneous Equipment	-	0	0	-	0	0
Air Internal Energy Change	-	0		-	0	0
Safety Factor	0% / 0%	0	0	0%	0	0
>> Total Zone Loads	**-**	**14537**	**5243**	**-**	**7919**	**0**
Key:	**Positive values are cooling loads Negative values are heating loads**			**Positive values are heating loads Negative values are cooling loads**		

Note 1: Surface convection line items show the combined effects of conductive heat gain to the surface and radiative heat gains absorbed at the surface which are then convected to room air.

Note 2: Lighting, equipment, and people line items include only the direct convective heat gain from the heat source to the room air. The radiative portion of the heat gain is first absorbed by surfaces in the room and then later convected from the surface to the air. Therefore the effect of the radiative portion of the heat gain is found in the surface convection line items.

Note 3: Solar heat gain is absorbed by surfaces in the room, re-radiated to other surfaces, and finally convected from the surfaces to room air. Therefore, the effect of solar heat gain is found in the surface convection line items.

Air System Sizing Summary for HP-107-BUFFET/SERVING

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton) - Copy(converted) 06/11/2026
Prepared by: Shakespeare Engineering 11:07 AM

Air System Information

Air System Name	HP-107-BUFFET/SERVING		Number of zones	1	
Equipment Class	SPLT AHU		Floor Area	747.0	sqft
Air System Type	SZCAV		Location	St George Municipal, UT, USA	

Sizing Calculation Information

Calculation Months	Jan to Dec	Zone CFM Sizing	Sum of space airflow rates
Sizing Data	Calculated	Space CFM Sizing	Individual peak space loads

Central Cooling Coil Sizing Data

Total coil load	1.9	Tons	Peak coil load occurs at	July 16:00	
Total coil load	23.3	MBH	OA DB / WB	106.2 / 66.2	F
Sensible coil load	23.3	MBH	Entering DB / WB	82.4 / 61.5	F
Coil CFM at peak load	779	CFM	Leaving DB / WB	52.1 / 50.3	F
Sum of peak zone CFM	779	CFM	Resulting RH	47	%
Sensible heat ratio	1.000		Design supply temp.	55.0	F
CFM/Ton	401.0		Zone T-stat Check	1 of 1	OK
sqft/Ton	384.3		Max zone temperature deviation	0.0	F
BTU/(hr sqft)	31.2				
Water flow @ 10.0 F rise	N/A				

Central Heating Coil Sizing Data

Max coil load	14.9	MBH	Load occurs at	Design Heating	
Coil CFM at Design Heating	779	CFM	BTU/(hr sqft)	20.0	
Max coil CFM	779	CFM	Ent. DB / Lvg DB	57.1 / 76.7	F
Water flow @ 20.0 F drop	N/A				

Supply Fan Sizing Data

Design CFM	779	CFM	Fan motor BHP	0.43	BHP
Design CFM/sqft	1.04	CFM/sqft	Fan motor kW	0.34	kW
			Fan total static	2.00	in wg

Outdoor Ventilation Air Data

Design airflow CFM	224	CFM	CFM/person	22.50	CFM/person
CFM/sqft	0.30	CFM/sqft			

Air System Information

Air System Name	**HP-107-BUFFET/SERVING**	Number of zones	**1**
Equipment Class	**SPLT AHU**	Floor Area	**747.0** sqft
Air System Type	**SZCAV**	Location	**St George Municipal, UT, USA**

Sizing Calculation Information

Calculation Months	**Jan to Dec**	Zone CFM Sizing	**Sum of space airflow rates**
Sizing Data	**Calculated**	Space CFM Sizing	**Individual peak space loads**

Zone Terminal Sizing Data

Zone Name	Design Supply Airflow (CFM)	Minimum Supply Airflow (CFM)	Zone CFM/sqft	Reheat Coil Capacity (MBH)	Reheat Coil Water gpm @ 20.0 F	Zone Htg Unit Coil Capacity (MBH)	Zone Htg Unit Water gpm @ 20.0 F	Mixing Box Fan Airflow (CFM)
HP-107-BUFFET/SERVING	779	779	1.04	0.0	-	0.0	-	0

Zone Peak Sensible Loads

Zone Name	Zone Cooling Sensible (MBH)	Time of Peak Sensible Cooling Load	Zone Heating Load (MBH)	Zone Floor Area (sqft)
HP-107-BUFFET/SERVING	13.6	July 18:00	3.5	747.0

Space Loads and Airflows

Zone Name / Space Name	Cooling Sensible (MBH)	Time of Peak Sensible Load	Air Flow (CFM)	Heating Load (MBH)	Floor Area (sqft)	Space CFM/sqft
HP-107-BUFFET/SERVING						
106-BREAKFAST/BUFFETT	13.6	July 18:00	779	3.5	747.0	1.04

Ventilation Sizing Summary for HP-107-BUFFET/SERVING
(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton) - Copy(converted) 06/11/2026
Prepared by: Shakespeare Engineering 11:07 AM

1. Summary

Ventilation Sizing Method .... **Sum of Space OA Airflows**
Design Ventilation Airflow Rate .... **164** CFM
Design Ventilation Airflow Rate, Corrected for Exhaust Air .... **224** CFM

2. Space Ventilation Analysis

Zone Name / Space Name	Floor Area (sqft)	Maximum Occupants	Maximum Supply Air (CFM)	Required Outdoor Air (CFM/person)	Required Outdoor Air (CFM/sqft)	Required Outdoor Air (CFM)	Required Outdoor Air (% of supply)	Required Outdoor Air (ACH)	Uncorrected Outdoor Air (CFM)	Direct Exhaust Air (CFM)
HP-107-BUFFET/SERVING										
106-BREAKFAST/BUFFETT	747.0	10.0	779.4	7.50	0.12	0.0	0.0	0.00	164.3	224.1
Totals	**747.0**	**10.0**	**779.4**						**164.3**	

Air System Heat Balance Summary for HP-107-BUFFET/SERVING

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton) - Copy(converted) 06/11/2026
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Table 1. System Loads

	DESIGN COOLING - JULY 16:00			DESIGN HEATING		
	OA DB / WB 106.2 F / 66.2 F			OA DB / WB 20.2 F / 16.3 F		
COMPONENT LOADS	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**
Zone Conditioning	-	14926	2588	-	4762	0
Plenum Load	-	0	0	-	0	0
Return Fan Load	555 CFM	0	-	555 CFM	0	-
Ventilation Load	224 CFM	7164	-2588	224 CFM	11262	0
Supply Fan Load	779 CFM	1155	-	779 CFM	-1155	-
Zone Fan Coil Fans Load	-	0	-	-	0	-
>> Total System Loads	**-**	**23246**	**0**	**-**	**14869**	**0**
Central Cooling Coil	-	23325	0	-	0	0
Central Heating Coil	-	0	-	-	14917	-
>> Total Conditioning	**-**	**23325**	**0**	**-**	**14917**	**0**
Key:	**Positive values are cooling loads Negative values are heating loads**			**Positive values are heating loads Negative values are cooling loads**		

Table 2. Zone Heat Balance Loads

	DESIGN COOLING - JULY 16:00			DESIGN HEATING		
	OA DB / WB 106.2 F / 66.2 F			OA DB / WB 20.2 F / 16.3 F		
Zone Heat Balance Component	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**
Exterior Wall Convection	275 sqft	1035	-	275 sqft	668	-
Roof Convection	0 sqft	0	-	0 sqft	0	-
Window Convection	0 sqft	0	-	0 sqft	0	-
Skylight Convection	0 sqft	0	-	0 sqft	0	-
Door Convection	0 sqft	0	-	0 sqft	0	-
Floor Convection	747 sqft	1839	-	747 sqft	1838	-
Interior Wall Convection	273 sqft	577	-	273 sqft	136	-
Ceiling Convection	747 sqft	2684	-	747 sqft	853	-
Overhead Lighting Convection	550 W	833	-	0 W	0	-
Task Lighting Convection	0 W	0	-	0 W	0	-
Electric Equipment Convection	2241 W	5735	-	0 W	0	-
People Convection	10	837	2689	0	0	0
Infiltration	0 CFM	0	0	0 CFM	0	0
Miscellaneous Equipment	-	0	0	-	0	0
Air Internal Energy Change	-	0		-	0	0
Safety Factor	0% / 0%	0	0	0%	0	0
>> Total Zone Loads	**-**	**13538**	**2689**	**-**	**3495**	**0**
Key:	**Positive values are cooling loads Negative values are heating loads**			**Positive values are heating loads Negative values are cooling loads**		

Note 1: Surface convection line items show the combined effects of conductive heat gain to the surface and radiative heat gains absorbed at the surface which are then convected to room air.

Note 2: Lighting, equipment, and people line items include only the direct convective heat gain from the heat source to the room air. The radiative portion of the heat gain is first absorbed by surfaces in the room and then later convected from the surface to the air. Therefore the effect of the radiative portion of the heat gain is found in the surface convection line items.

Note 3: Solar heat gain is absorbed by surfaces in the room, re-radiated to other surfaces, and finally convected from the surfaces to room air. Therefore, the effect of solar heat gain is found in the surface convection line items.

Air System Information

Air System Name	**HP-108-FOOD PREP**		Number of zones	**1**	
Equipment Class	**SPLT AHU**		Floor Area	**695.6**	sqft
Air System Type	**SZCAV**		Location	**St George Municipal, UT, USA**	

Sizing Calculation Information

Calculation Months	**Jan to Dec**	Zone CFM Sizing	**Sum of space airflow rates**
Sizing Data	**Calculated**	Space CFM Sizing	**Individual peak space loads**

Central Cooling Coil Sizing Data

Total coil load	**3.0**	Tons	Peak coil load occurs at	**July 16:00**	
Total coil load	**36.6**	MBH	OA DB / WB	**106.2 / 66.2**	F
Sensible coil load	**36.6**	MBH	Entering DB / WB	**87.8 / 61.9**	F
Coil CFM at peak load	**1080**	CFM	Leaving DB / WB	**53.5 / 49.1**	F
Sum of peak zone CFM	**1080**	CFM	Resulting RH	**42**	%
Sensible heat ratio	**1.000**		Design supply temp.	**55.0**	F
CFM/Ton	**354.5**		Zone T-stat Check	**1 of 1**	OK
sqft/Ton	**228.3**		Max zone temperature deviation	**0.0**	F
BTU/(hr sqft)	**52.6**				
Water flow @ 10.0 F rise	**N/A**				

Central Heating Coil Sizing Data

Max coil load	**26.6**	MBH	Load occurs at	**Design Heating**	
Coil CFM at Design Heating	**1080**	CFM	BTU/(hr sqft)	**38.2**	
Max coil CFM	**1080**	CFM	Ent. DB / Lvg DB	**48.6 / 73.9**	F
Water flow @ 20.0 F drop	**N/A**				

Supply Fan Sizing Data

Design CFM	**1080**	CFM	Fan motor BHP	**0.59**	BHP
Design CFM/sqft	**1.55**	CFM/sqft	Fan motor kW	**0.47**	kW
			Fan total static	**2.00**	in wg

Outdoor Ventilation Air Data

Design airflow CFM	**487**	CFM	CFM/person	**35.00**	CFM/person
CFM/sqft	**0.70**	CFM/sqft			

Zone Sizing Summary for HP-108-FOOD PREP

(In Alternative: Default Alternative)

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Air System Information

Air System Name	**HP-108-FOOD PREP**	Number of zones	**1**
Equipment Class	**SPLT AHU**	Floor Area	**695.6** sqft
Air System Type	**SZCAV**	Location	**St George Municipal, UT, USA**

Sizing Calculation Information

Calculation Months	**Jan to Dec**	Zone CFM Sizing	**Sum of space airflow rates**
Sizing Data	**Calculated**	Space CFM Sizing	**Individual peak space loads**

Zone Terminal Sizing Data

Zone Name	Design Supply Airflow (CFM)	Minimum Supply Airflow (CFM)	Zone CFM/sqft	Reheat Coil Capacity (MBH)	Reheat Coil Water gpm @ 20.0 F	Zone Htg Unit Coil Capacity (MBH)	Zone Htg Unit Water gpm @ 20.0 F	Mixing Box Fan Airflow (CFM)
HP-108-FOOD PREP	1080	1080	1.55	0.0	-	0.0	-	0

Zone Peak Sensible Loads

Zone Name	Zone Cooling Sensible (MBH)	Time of Peak Sensible Cooling Load	Zone Heating Load (MBH)	Zone Floor Area (sqft)
HP-108-FOOD PREP	18.9	July 20:00	3.1	695.6

Space Loads and Airflows

Zone Name / Space Name	Cooling Sensible (MBH)	Time of Peak Sensible Load	Air Flow (CFM)	Heating Load (MBH)	Floor Area (sqft)	Space CFM/sqft
HP-108-FOOD PREP						
107-FOOD PREP	18.9	July 20:00	1080	3.1	695.6	1.55

Ventilation Sizing Summary for HP-108-FOOD PREP
(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton) - Copy(converted) 06/11/2026
Prepared by: Shakespeare Engineering 11:07 AM

1. Summary

Ventilation Sizing Method .... **Sum of Space OA Airflows**
Design Ventilation Airflow Rate .... **188** CFM
Design Ventilation Airflow Rate, Corrected for Exhaust Air .... **487** CFM

2. Space Ventilation Analysis

Zone Name / Space Name	Floor Area (sqft)	Maximum Occupants	Maximum Supply Air (CFM)	Required Outdoor Air (CFM/person)	Required Outdoor Air (CFM/sqft)	Required Outdoor Air (CFM)	Required Outdoor Air (% of supply)	Required Outdoor Air (ACH)	Uncorrected Outdoor Air (CFM)	Direct Exhaust Air (CFM)
HP-108-FOOD PREP										
107-FOOD PREP	695.6	13.9	1079.9	7.50	0.12	0.0	0.0	0.00	187.8	486.9
Totals	**695.6**	**13.9**	**1079.9**						**187.8**	

Table 1. System Loads

	DESIGN COOLING - JULY 16:00			DESIGN HEATING		
	OA DB / WB 106.2 F / 66.2 F			OA DB / WB 20.2 F / 16.3 F		
COMPONENT LOADS	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**
Zone Conditioning	-	19202	3615	-	3621	0
Plenum Load	-	0	0	-	0	0
Return Fan Load	593 CFM	0	-	593 CFM	0	-
Ventilation Load	487 CFM	15567	-3615	487 CFM	24470	0
Supply Fan Load	1080 CFM	1601	-	1080 CFM	-1601	-
Zone Fan Coil Fans Load	-	0	-	-	0	-
>> Total System Loads	**-**	**36370**	**0**	**-**	**26490**	**0**
Central Cooling Coil	-	36559	0	-	0	0
Central Heating Coil	-	0	-	-	26594	-
>> Total Conditioning	**-**	**36559**	**0**	**-**	**26594**	**0**
Key:	**Positive values are cooling loads Negative values are heating loads**			**Positive values are heating loads Negative values are cooling loads**		

Table 2. Zone Heat Balance Loads

	DESIGN COOLING - JULY 16:00			DESIGN HEATING		
	OA DB / WB 106.2 F / 66.2 F			OA DB / WB 20.2 F / 16.3 F		
Zone Heat Balance Component	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**
Exterior Wall Convection	314 sqft	1360	-	314 sqft	677	-
Roof Convection	0 sqft	0	-	0 sqft	0	-
Window Convection	0 sqft	0	-	0 sqft	0	-
Skylight Convection	0 sqft	0	-	0 sqft	0	-
Door Convection	0 sqft	0	-	0 sqft	0	-
Floor Convection	696 sqft	1905	-	696 sqft	1633	-
Interior Wall Convection	741 sqft	1824	-	741 sqft	228	-
Ceiling Convection	696 sqft	2808	-	696 sqft	572	-
Overhead Lighting Convection	512 W	775	-	0 W	0	-
Task Lighting Convection	0 W	0	-	0 W	0	-
Electric Equipment Convection	3478 W	8900	-	0 W	0	-
People Convection	14	1169	3756	0	0	0
Infiltration	0 CFM	0	0	0 CFM	0	0
Miscellaneous Equipment	-	0	0	-	0	0
Air Internal Energy Change	-	0		-	0	0
Safety Factor	0% / 0%	0	0	0%	0	0
>> Total Zone Loads	**-**	**18741**	**3756**	**-**	**3110**	**0**
Key:	**Positive values are cooling loads Negative values are heating loads**			**Positive values are heating loads Negative values are cooling loads**		

Note 1: Surface convection line items show the combined effects of conductive heat gain to the surface and radiative heat gains absorbed at the surface which are then convected to room air.

Note 2: Lighting, equipment, and people line items include only the direct convective heat gain from the heat source to the room air. The radiative portion of the heat gain is first absorbed by surfaces in the room and then later convected from the surface to the air. Therefore the effect of the radiative portion of the heat gain is found in the surface convection line items.

Note 3: Solar heat gain is absorbed by surfaces in the room, re-radiated to other surfaces, and finally convected from the surfaces to room air. Therefore, the effect of solar heat gain is found in the surface convection line items.

Air System Sizing Summary for HP-109-DINING FLEX

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton) - Copy(converted) 06/11/2026
Prepared by: Shakespeare Engineering 11:07 AM

Air System Information

Air System Name	**HP-109-DINING FLEX**		Number of zones	**1**	
Equipment Class	**SPLT AHU**		Floor Area	**362.7**	sqft
Air System Type	**SZCAV**		Location	**St George Municipal, UT, USA**	

Sizing Calculation Information

Calculation Months	**Jan to Dec**	Zone CFM Sizing	**Sum of space airflow rates**
Sizing Data	**Calculated**	Space CFM Sizing	**Individual peak space loads**

Central Cooling Coil Sizing Data

Total coil load	**1.1**	Tons	Peak coil load occurs at	**August 15:00**	
Total coil load	**12.9**	MBH	OA DB / WB	**106.2 / 66.2**	F
Sensible coil load	**12.9**	MBH	Entering DB / WB	**88.9 / 62.5**	F
Coil CFM at peak load	**361**	CFM	Leaving DB / WB	**52.8 / 49.2**	F
Sum of peak zone CFM	**361**	CFM	Resulting RH	**44**	%
Sensible heat ratio	**1.000**		Design supply temp.	**55.0**	F
CFM/Ton	**337.4**		Zone T-stat Check	**1 of 1**	OK
sqft/Ton	**338.7**		Max zone temperature deviation	**0.0**	F
BTU/(hr sqft)	**35.4**				
Water flow @ 10.0 F rise	**N/A**				

Central Heating Coil Sizing Data

Max coil load	**11.4**	MBH	Load occurs at	**Design Heating**	
Coil CFM at Design Heating	**361**	CFM	BTU/(hr sqft)	**31.5**	
Max coil CFM	**361**	CFM	Ent. DB / Lvg DB	**47.0 / 79.4**	F
Water flow @ 20.0 F drop	**N/A**				

Supply Fan Sizing Data

Design CFM	**361**	CFM	Fan motor BHP	**0.20**	BHP
Design CFM/sqft	**1.00**	CFM/sqft	Fan motor kW	**0.16**	kW
			Fan total static	**2.00**	in wg

Outdoor Ventilation Air Data

Design airflow CFM	**174**	CFM	CFM/person	**12.00**	CFM/person
CFM/sqft	**0.48**	CFM/sqft			

Zone Sizing Summary for HP-109-DINING FLEX
(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton) - Copy(converted) 06/11/2026
Prepared by: Shakespeare Engineering 11:07 AM

Air System Information

Air System Name ..... **HP-109-DINING FLEX**
Equipment Class ..... **SPLT AHU**
Air System Type ..... **SZCAV**

Number of zones **1**
Floor Area ..... **362.7** sqft
Location ..... **St George Municipal, UT, USA**

Sizing Calculation Information

Calculation Months ..... **Jan to Dec**
Sizing Data ..... **Calculated**

Zone CFM Sizing ..... **Sum of space airflow rates**
Space CFM Sizing ..... **Individual peak space loads**

Zone Terminal Sizing Data

Zone Name	Design Supply Airflow (CFM)	Minimum Supply Airflow (CFM)	Zone CFM/sqft	Reheat Coil Capacity (MBH)	Reheat Coil Water gpm @ 20.0 F	Zone Htg Unit Coil Capacity (MBH)	Zone Htg Unit Water gpm @ 20.0 F	Mixing Box Fan Airflow (CFM)
HP-109-DINING FLEX	361	361	1.00	0.0	-	0.0	-	0

Zone Peak Sensible Loads

Zone Name	Zone Cooling Sensible (MBH)	Time of Peak Sensible Cooling Load	Zone Heating Load (MBH)	Zone Floor Area (sqft)
HP-109-DINING FLEX	6.3	September 13:00	2.5	362.7

Space Loads and Airflows

Zone Name / Space Name	Cooling Sensible (MBH)	Time of Peak Sensible Load	Air Flow (CFM)	Heating Load (MBH)	Floor Area (sqft)	Space CFM/sqft
HP-109-DINING FLEX						
109-DINING/FLEX	6.3	September 13:00	361	2.5	362.7	1.00

Ventilation Sizing Summary for HP-109-DINING FLEX

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton) - Copy(converted) 06/11/2026
Prepared by: Shakespeare Engineering 11:07 AM

1. Summary

Ventilation Sizing Method .... **Sum of Space OA Airflows**
Design Ventilation Airflow Rate .... **174** CFM

2. Space Ventilation Analysis

Zone Name / Space Name	Floor Area (sqft)	Maximum Occupants	Maximum Supply Air (CFM)	Required Outdoor Air (CFM/person)	Required Outdoor Air (CFM/sqft)	Required Outdoor Air (CFM)	Required Outdoor Air (% of supply)	Required Outdoor Air (ACH)	Uncorrected Outdoor Air (CFM)	Direct Exhaust Air (CFM)
HP-109-DINING FLEX										
109-DINING/FLEX	362.7	14.5	361.3	7.50	0.18	0.0	0.0	0.00	174.1	0.0
Totals	**362.7**	**14.5**	**361.3**						**174.1**	

Air System Heat Balance Summary for HP-109-DINING FLEX

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton) - Copy(converted) 06/11/2026
Prepared by: Shakespeare Engineering 11:07 AM

Table 1. System Loads

	DESIGN COOLING - AUGUST 15:00			DESIGN HEATING		
	OA DB / WB 106.2 F / 66.2 F			OA DB / WB 20.2 F / 16.3 F		
COMPONENT LOADS	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**
Zone Conditioning	-	6661	1675	-	3157	0
Plenum Load	-	0	0	-	0	0
Return Fan Load	361 CFM	0	-	361 CFM	0	-
Ventilation Load	174 CFM	5591	-1675	174 CFM	8750	0
Supply Fan Load	361 CFM	536	-	361 CFM	-536	-
Zone Fan Coil Fans Load	-	0	-	-	0	-
>> Total System Loads	**-**	**12788**	**0**	**-**	**11372**	**0**
Central Cooling Coil	-	12852	0	-	0	0
Central Heating Coil	-	0	-	-	11410	-
>> Total Conditioning	**-**	**12852**	**0**	**-**	**11410**	**0**
Key:	**Positive values are cooling loads Negative values are heating loads**			**Positive values are heating loads Negative values are cooling loads**		

Table 2. Zone Heat Balance Loads

	DESIGN COOLING - AUGUST 15:00			DESIGN HEATING		
	OA DB / WB 106.2 F / 66.2 F			OA DB / WB 20.2 F / 16.3 F		
Zone Heat Balance Component	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**
Exterior Wall Convection	92 sqft	446	-	92 sqft	239	-
Roof Convection	0 sqft	0	-	0 sqft	0	-
Window Convection	58 sqft	500	-	58 sqft	484	-
Skylight Convection	0 sqft	0	-	0 sqft	0	-
Door Convection	0 sqft	0	-	0 sqft	0	-
Floor Convection	363 sqft	942	-	363 sqft	1014	-
Interior Wall Convection	485 sqft	997	-	485 sqft	294	-
Ceiling Convection	363 sqft	1295	-	363 sqft	502	-
Overhead Lighting Convection	105 W	160	-	0 W	0	-
Task Lighting Convection	0 W	0	-	0 W	0	-
Electric Equipment Convection	181 W	464	-	0 W	0	-
People Convection	15	1001	1741	0	0	0
Infiltration	0 CFM	0	0	0 CFM	0	0
Miscellaneous Equipment	-	0	0	-	0	0
Air Internal Energy Change	-	0		-	0	0
Safety Factor	0% / 0%	0	0	0%	0	0
>> Total Zone Loads	**-**	**5805**	**1741**	**-**	**2532**	**0**
Key:	**Positive values are cooling loads Negative values are heating loads**			**Positive values are heating loads Negative values are cooling loads**		

Note 1: Surface convection line items show the combined effects of conductive heat gain to the surface and radiative heat gains absorbed at the surface which are then convected to room air.

Note 2: Lighting, equipment, and people line items include only the direct convective heat gain from the heat source to the room air. The radiative portion of the heat gain is first absorbed by surfaces in the room and then later convected from the surface to the air. Therefore the effect of the radiative portion of the heat gain is found in the surface convection line items.

Note 3: Solar heat gain is absorbed by surfaces in the room, re-radiated to other surfaces, and finally convected from the surfaces to room air. Therefore, the effect of solar heat gain is found in the surface convection line items.

Air System Information

Air System Name	**HP-110-DINING SMALL**		Number of zones	**1**	
Equipment Class	**SPLT AHU**		Floor Area	**713.0**	sqft
Air System Type	**SZCAV**		Location	**St George Municipal, UT, USA**	

Sizing Calculation Information

Calculation Months	**Jan to Dec**	Zone CFM Sizing	**Sum of space airflow rates**
Sizing Data	**Calculated**	Space CFM Sizing	**Individual peak space loads**

Central Cooling Coil Sizing Data

Total coil load	**2.0**	Tons	Peak coil load occurs at	**August 14:00**	
Total coil load	**24.5**	MBH	OA DB / WB	**104.9 / 65.8**	F
Sensible coil load	**24.5**	MBH	Entering DB / WB	**88.5 / 62.4**	F
Coil CFM at peak load	**703**	CFM	Leaving DB / WB	**53.2 / 49.3**	F
Sum of peak zone CFM	**703**	CFM	Resulting RH	**44**	%
Sensible heat ratio	**1.000**		Design supply temp.	**55.0**	F
CFM/Ton	**344.1**		Zone T-stat Check	**1 of 1**	OK
sqft/Ton	**349.1**		Max zone temperature deviation	**0.0**	F
BTU/(hr sqft)	**34.4**				
Water flow @ 10.0 F rise	**N/A**				

Central Heating Coil Sizing Data

Max coil load	**22.4**	MBH	Load occurs at	**Design Heating**	
Coil CFM at Design Heating	**703**	CFM	BTU/(hr sqft)	**31.4**	
Max coil CFM	**703**	CFM	Ent. DB / Lvg DB	**46.8 / 79.5**	F
Water flow @ 20.0 F drop	**N/A**				

Supply Fan Sizing Data

Design CFM	**703**	CFM	Fan motor BHP	**0.38**	BHP
Design CFM/sqft	**0.99**	CFM/sqft	Fan motor kW	**0.31**	kW
			Fan total static	**2.00**	in wg

Outdoor Ventilation Air Data

Design airflow CFM	**342**	CFM	CFM/person	**12.00**	CFM/person
CFM/sqft	**0.48**	CFM/sqft			

Air System Information

Air System Name .... **HP-110-DINING SMALL**
Equipment Class .... **SPLT AHU**
Air System Type .... **SZCAV**

Number of zones **1**
Floor Area .... **713.0** sqft
Location .... **St George Municipal, UT, USA**

Sizing Calculation Information

Calculation Months .... **Jan to Dec**
Sizing Data .... **Calculated**

Zone CFM Sizing .... **Sum of space airflow rates**
Space CFM Sizing .... **Individual peak space loads**

Zone Terminal Sizing Data

Zone Name	Design Supply Airflow (CFM)	Minimum Supply Airflow (CFM)	Zone CFM/sqft	Reheat Coil Capacity (MBH)	Reheat Coil Water gpm @ 20.0 F	Zone Htg Unit Coil Capacity (MBH)	Zone Htg Unit Water gpm @ 20.0 F	Mixing Box Fan Airflow (CFM)
HP-110-DINING SMALL	703	703	0.99	0.0	-	0.0	-	0

Zone Peak Sensible Loads

Zone Name	Zone Cooling Sensible (MBH)	Time of Peak Sensible Cooling Load	Zone Heating Load (MBH)	Zone Floor Area (sqft)
HP-110-DINING SMALL	12.3	October 13:00	5.1	713.0

Space Loads and Airflows

Zone Name / Space Name	Cooling Sensible (MBH)	Time of Peak Sensible Load	Air Flow (CFM)	Heating Load (MBH)	Floor Area (sqft)	Space CFM/sqft
HP-110-DINING SMALL						
105B - DINING	12.3	October 13:00	703	5.1	713.0	0.99

1. Summary

Ventilation Sizing Method .... **Sum of Space OA Airflows**
Design Ventilation Airflow Rate .... **342** CFM

2. Space Ventilation Analysis

Zone Name / Space Name	Floor Area (sqft)	Maximum Occupants	Maximum Supply Air (CFM)	Required Outdoor Air (CFM/person)	Required Outdoor Air (CFM/sqft)	Required Outdoor Air (CFM)	Required Outdoor Air (% of supply)	Required Outdoor Air (ACH)	Uncorrected Outdoor Air (CFM)	Direct Exhaust Air (CFM)
HP-110-DINING SMALL										
105B - DINING	713.0	28.5	702.6	7.50	0.18	0.0	0.0	0.00	342.2	0.0
Totals	**713.0**	**28.5**	**702.6**						**342.2**	

Air System Heat Balance Summary for HP-110-DINING SMALL

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton) - Copy(converted) 06/11/2026

Prepared by: Shakespeare Engineering 11:07 AM

Table 1. System Loads

	DESIGN COOLING - AUGUST 14:00			DESIGN HEATING		
	OA DB / WB 104.9 F / 65.8 F			OA DB / WB 20.2 F / 16.3 F		
COMPONENT LOADS	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**
Zone Conditioning	-	12724	3291	-	6174	0
Plenum Load	-	0	0	-	0	0
Return Fan Load	703 CFM	0	-	703 CFM	0	-
Ventilation Load	342 CFM	10618	-3287	342 CFM	17199	0
Supply Fan Load	703 CFM	1042	-	703 CFM	-1042	-
Zone Fan Coil Fans Load	-	0	-	-	0	-
>> Total System Loads	**-**	**24384**	**3**	**-**	**22331**	**0**
Central Cooling Coil	-	24506	0	-	0	0
Central Heating Coil	-	0	-	-	22405	-
>> Total Conditioning	**-**	**24506**	**0**	**-**	**22405**	**0**
Key:	**Positive values are cooling loads Negative values are heating loads**			**Positive values are heating loads Negative values are cooling loads**		

Table 2. Zone Heat Balance Loads

	DESIGN COOLING - AUGUST 14:00			DESIGN HEATING		
	OA DB / WB 104.9 F / 65.8 F			OA DB / WB 20.2 F / 16.3 F		
Zone Heat Balance Component	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**
Exterior Wall Convection	98 sqft	449	-	98 sqft	257	-
Roof Convection	0 sqft	0	-	0 sqft	0	-
Window Convection	196 sqft	2717	-	196 sqft	1662	-
Skylight Convection	0 sqft	0	-	0 sqft	0	-
Door Convection	0 sqft	0	-	0 sqft	0	-
Floor Convection	713 sqft	2053	-	713 sqft	1998	-
Interior Wall Convection	242 sqft	507	-	242 sqft	147	-
Ceiling Convection	713 sqft	2557	-	713 sqft	993	-
Overhead Lighting Convection	207 W	314	-	0 W	0	-
Task Lighting Convection	0 W	0	-	0 W	0	-
Electric Equipment Convection	356 W	912	-	0 W	0	-
People Convection	29	1968	3422	0	0	0
Infiltration	0 CFM	0	0	0 CFM	0	0
Miscellaneous Equipment	-	0	0	-	0	0
Air Internal Energy Change	-	0		-	0	0
Safety Factor	0% / 0%	0	0	0%	0	0
>> Total Zone Loads	**-**	**11477**	**3422**	**-**	**5058**	**0**
Key:	**Positive values are cooling loads Negative values are heating loads**			**Positive values are heating loads Negative values are cooling loads**		

Note 1: Surface convection line items show the combined effects of conductive heat gain to the surface and radiative heat gains absorbed at the surface which are then convected to room air.

Note 2: Lighting, equipment, and people line items include only the direct convective heat gain from the heat source to the room air. The radiative portion of the heat gain is first absorbed by surfaces in the room and then later convected from the surface to the air. Therefore the effect of the radiative portion of the heat gain is found in the surface convection line items.

Note 3: Solar heat gain is absorbed by surfaces in the room, re-radiated to other surfaces, and finally convected from the surfaces to room air. Therefore, the effect of solar heat gain is found in the surface convection line items.

Air System Sizing Summary for HP-111-MEETING

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton) - Copy(converted)	06/11/2026
Prepared by: Shakespeare Engineering	11:07 AM

Air System Information

Air System Name	**HP-111-MEETING**		Number of zones	**1**	
Equipment Class	**SPLT AHU**		Floor Area	**808.8**	sqft
Air System Type	**SZCAV**		Location	**St George Municipal, UT, USA**	

Sizing Calculation Information

Calculation Months	**Jan to Dec**	Zone CFM Sizing	**Sum of space airflow rates**
Sizing Data	**Calculated**	Space CFM Sizing	**Individual peak space loads**

Central Cooling Coil Sizing Data

Total coil load	**2.4**	Tons	Peak coil load occurs at	**September 14:00**	
Total coil load	**29.4**	MBH	OA DB / WB	**99.7 / 62.8**	F
Sensible coil load	**28.4**	MBH	Entering DB / WB	**76.7 / 62.3**	F
Coil CFM at peak load	**1281**	CFM	Leaving DB / WB	**54.3 / 54.1**	F
Sum of peak zone CFM	**1281**	CFM	Resulting RH	**56**	%
Sensible heat ratio	**0.968**		Design supply temp.	**55.0**	F
CFM/Ton	**523.6**		Zone T-stat Check	**1 of 1**	OK
sqft/Ton	**330.6**		Max zone temperature deviation	**0.0**	F
BTU/(hr sqft)	**36.3**				
Water flow @ 10.0 F rise	**N/A**				

Central Heating Coil Sizing Data

Max coil load	**15.3**	MBH	Load occurs at	**Design Heating**	
Coil CFM at Design Heating	**1281**	CFM	BTU/(hr sqft)	**18.9**	
Max coil CFM	**1281**	CFM	Ent. DB / Lvg DB	**64.7 / 77.0**	F
Water flow @ 20.0 F drop	**N/A**				

Supply Fan Sizing Data

Design CFM	**1281**	CFM	Fan motor BHP	**0.70**	BHP
Design CFM/sqft	**1.58**	CFM/sqft	Fan motor kW	**0.56**	kW
			Fan total static	**2.00**	in wg

Outdoor Ventilation Air Data

Design airflow CFM	**180**	CFM	CFM/person	**6.84**	CFM/person
CFM/sqft	**0.22**	CFM/sqft			

Air System Information

Air System Name	**HP-111-MEETING**	Number of zones	**1**
Equipment Class	**SPLT AHU**	Floor Area	**808.8** sqft
Air System Type	**SZCAV**	Location	**St George Municipal, UT, USA**

Sizing Calculation Information

Calculation Months	**Jan to Dec**	Zone CFM Sizing	**Sum of space airflow rates**
Sizing Data	**Calculated**	Space CFM Sizing	**Individual peak space loads**

Zone Terminal Sizing Data

Zone Name	Design Supply Airflow (CFM)	Minimum Supply Airflow (CFM)	Zone CFM/sqft	Reheat Coil Capacity (MBH)	Reheat Coil Water gpm @ 20.0 F	Zone Htg Unit Coil Capacity (MBH)	Zone Htg Unit Water gpm @ 20.0 F	Mixing Box Fan Airflow (CFM)
HP-111-MEETING	1281	1281	1.58	0.0	-	0.0	-	0

Zone Peak Sensible Loads

Zone Name	Zone Cooling Sensible (MBH)	Time of Peak Sensible Cooling Load	Zone Heating Load (MBH)	Zone Floor Area (sqft)
HP-111-MEETING	22.1	November 13:00	7.1	808.8

Space Loads and Airflows

Zone Name / Space Name	Cooling Sensible (MBH)	Time of Peak Sensible Load	Air Flow (CFM)	Heating Load (MBH)	Floor Area (sqft)	Space CFM/sqft
HP-111-MEETING						
111-MEETING ROOM	20.0	November 13:00	1142	5.5	658.9	1.73
111-MEETING STORAGE	2.4	September 16:00	138	1.6	149.9	0.92

Ventilation Sizing Summary for HP-111-MEETING
(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton) - Copy(converted) 06/11/2026
Prepared by: Shakespeare Engineering 11:07 AM

1. Summary

Ventilation Sizing Method .... **Sum of Space OA Airflows**
Design Ventilation Airflow Rate .... **180** CFM

2. Space Ventilation Analysis

Zone Name / Space Name	Floor Area (sqft)	Maximum Occupants	Maximum Supply Air (CFM)	Required Outdoor Air (CFM/person)	Required Outdoor Air (CFM/sqft)	Required Outdoor Air (CFM)	Required Outdoor Air (% of supply)	Required Outdoor Air (ACH)	Uncorrected Outdoor Air (CFM)	Direct Exhaust Air (CFM)
HP-111-MEETING										
111-MEETING ROOM	658.9	26.4	1142.5	5.00	0.06	0.0	0.0	0.00	171.3	0.0
111-MEETING STORAGE	149.9	0.0	138.2	5.00	0.06	0.0	0.0	0.00	9.0	0.0
Totals	**808.8**	**26.4**	**1280.7**						**180.3**	

Air System Heat Balance Summary for HP-111-MEETING

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton) - Copy(converted) 06/11/2026
Prepared by: Shakespeare Engineering 11:07 AM

Table 1. System Loads

	DESIGN COOLING - SEPTEMBER 14:00			DESIGN HEATING		
	OA DB / WB 99.7 F / 62.8 F			OA DB / WB 20.2 F / 16.3 F		
COMPONENT LOADS	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**
Zone Conditioning	-	21793	5216	-	8117	0
Plenum Load	-	0	0	-	0	0
Return Fan Load	1281 CFM	0	-	1281 CFM	0	-
Ventilation Load	180 CFM	4685	-4334	180 CFM	9062	0
Supply Fan Load	1281 CFM	1899	-	1281 CFM	-1899	-
Zone Fan Coil Fans Load	-	0	-	-	0	-
>> Total System Loads	**-**	**28376**	**882**	**-**	**15280**	**0**
Central Cooling Coil	-	28402	953	-	0	0
Central Heating Coil	-	0	-	-	15319	-
>> Total Conditioning	**-**	**28402**	**953**	**-**	**15319**	**0**
Key:	**Positive values are cooling loads Negative values are heating loads**			**Positive values are heating loads Negative values are cooling loads**		

Table 2. Zone Heat Balance Loads

	DESIGN COOLING - SEPTEMBER 14:00			DESIGN HEATING		
	OA DB / WB 99.7 F / 62.8 F			OA DB / WB 20.2 F / 16.3 F		
Zone Heat Balance Component	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**
Exterior Wall Convection	244 sqft	1436	-	244 sqft	694	-
Roof Convection	0 sqft	0	-	0 sqft	0	-
Window Convection	184 sqft	3634	-	184 sqft	1579	-
Skylight Convection	0 sqft	0	-	0 sqft	0	-
Door Convection	0 sqft	0	-	0 sqft	0	-
Floor Convection	809 sqft	3127	-	809 sqft	2649	-
Interior Wall Convection	1106 sqft	4421	-	1106 sqft	843	-
Ceiling Convection	809 sqft	4888	-	809 sqft	1310	-
Overhead Lighting Convection	483 W	732	-	0 W	0	-
Task Lighting Convection	0 W	0	-	0 W	0	-
Electric Equipment Convection	329 W	843	-	0 W	0	-
People Convection	26	1937	5403	0	0	0
Infiltration	0 CFM	0	0	0 CFM	0	0
Miscellaneous Equipment	-	0	0	-	0	0
Air Internal Energy Change	-	0		-	0	0
Safety Factor	0% / 0%	0	0	0%	0	0
>> Total Zone Loads	**-**	**21018**	**5403**	**-**	**7075**	**0**
Key:	**Positive values are cooling loads Negative values are heating loads**			**Positive values are heating loads Negative values are cooling loads**		

Note 1: Surface convection line items show the combined effects of conductive heat gain to the surface and radiative heat gains absorbed at the surface which are then convected to room air.

Note 2: Lighting, equipment, and people line items include only the direct convective heat gain from the heat source to the room air. The radiative portion of the heat gain is first absorbed by surfaces in the room and then later convected from the surface to the air. Therefore the effect of the radiative portion of the heat gain is found in the surface convection line items.

Note 3: Solar heat gain is absorbed by surfaces in the room, re-radiated to other surfaces, and finally convected from the surfaces to room air. Therefore, the effect of solar heat gain is found in the surface convection line items.

Air System Sizing Summary for HP-112-ENGINEERING

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton) - Copy(converted) 06/11/2026
Prepared by: Shakespeare Engineering 11:07 AM

Air System Information

Air System Name	**HP-112-ENGINEERING**		Number of zones	**1**	
Equipment Class	**SPLT AHU**		Floor Area	**232.6**	sqft
Air System Type	**SZCAV**		Location	**St George Municipal, UT, USA**	

Sizing Calculation Information

Calculation Months	**Jan to Dec**	Zone CFM Sizing	**Sum of space airflow rates**
Sizing Data	**Calculated**	Space CFM Sizing	**Individual peak space loads**

Central Cooling Coil Sizing Data

Total coil load	**0.2**	Tons	Peak coil load occurs at	**July 16:00**	
Total coil load	**2.4**	MBH	OA DB / WB	**106.2 / 66.2**	F
Sensible coil load	**2.4**	MBH	Entering DB / WB	**82.9 / 62.5**	F
Coil CFM at peak load	**77**	CFM	Leaving DB / WB	**51.4 / 51.1**	F
Sum of peak zone CFM	**77**	CFM	Resulting RH	**52**	%
Sensible heat ratio	**1.000**		Design supply temp.	**55.0**	F
CFM/Ton	**385.9**		Zone T-stat Check	**1 of 1**	OK
sqft/Ton	**1160.3**		Max zone temperature deviation	**0.0**	F
BTU/(hr sqft)	**10.3**				
Water flow @ 10.0 F rise	**N/A**				

Central Heating Coil Sizing Data

Max coil load	**1.5**	MBH	Load occurs at	**Design Heating**	
Coil CFM at Design Heating	**77**	CFM	BTU/(hr sqft)	**6.5**	
Max coil CFM	**77**	CFM	Ent. DB / Lvg DB	**56.4 / 76.6**	F
Water flow @ 20.0 F drop	**N/A**				

Supply Fan Sizing Data

Design CFM	**77**	CFM	Fan motor BHP	**0.04**	BHP
Design CFM/sqft	**0.33**	CFM/sqft	Fan motor kW	**0.03**	kW
			Fan total static	**2.00**	in wg

Outdoor Ventilation Air Data

Design airflow CFM	**23**	CFM	CFM/person	**12.50**	CFM/person
CFM/sqft	**0.10**	CFM/sqft			

Air System Information

Air System Name	**HP-112-ENGINEERING**	Number of zones	**1**
Equipment Class	**SPLT AHU**	Floor Area	**232.6** sqft
Air System Type	**SZCAV**	Location	**St George Municipal, UT, USA**

Sizing Calculation Information

Calculation Months	**Jan to Dec**	Zone CFM Sizing	**Sum of space airflow rates**
Sizing Data	**Calculated**	Space CFM Sizing	**Individual peak space loads**

Zone Terminal Sizing Data

Zone Name	Design Supply Airflow (CFM)	Minimum Supply Airflow (CFM)	Zone CFM/sqft	Reheat Coil Capacity (MBH)	Reheat Coil Water gpm @ 20.0 F	Zone Htg Unit Coil Capacity (MBH)	Zone Htg Unit Water gpm @ 20.0 F	Mixing Box Fan Airflow (CFM)
HP-112-ENGINEERING	77	77	0.33	0.0	-	0.0	-	0

Zone Peak Sensible Loads

Zone Name	Zone Cooling Sensible (MBH)	Time of Peak Sensible Cooling Load	Zone Heating Load (MBH)	Zone Floor Area (sqft)
HP-112-ENGINEERING	1.4	July 19:00	0.4	232.6

Space Loads and Airflows

Zone Name / Space Name	Cooling Sensible (MBH)	Time of Peak Sensible Load	Air Flow (CFM)	Heating Load (MBH)	Floor Area (sqft)	Space CFM/sqft
HP-112-ENGINEERING						
134-ENGINEERING	1.4	July 19:00	77	0.4	232.6	0.33

Ventilation Sizing Summary for HP-112-ENGINEERING

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton) - Copy(converted) 06/11/2026
Prepared by: Shakespeare Engineering 11:07 AM

1. Summary

Ventilation Sizing Method .... **Sum of Space OA Airflows**
Design Ventilation Airflow Rate .... **23** CFM

2. Space Ventilation Analysis

Zone Name / Space Name	Floor Area (sqft)	Maximum Occupants	Maximum Supply Air (CFM)	Required Outdoor Air (CFM/person)	Required Outdoor Air (CFM/sqft)	Required Outdoor Air (CFM)	Required Outdoor Air (% of supply)	Required Outdoor Air (ACH)	Uncorrected Outdoor Air (CFM)	Direct Exhaust Air (CFM)
HP-112-ENGINEERING										
134-ENGINEERING	232.6	1.9	77.3	5.00	0.06	0.0	0.0	0.00	23.3	0.0
Totals	**232.6**	**1.9**	**77.3**						**23.3**	

Table 1. System Loads

	DESIGN COOLING - JULY 16:00			DESIGN HEATING		
	OA DB / WB 106.2 F / 66.2 F			OA DB / WB 20.2 F / 16.3 F		
COMPONENT LOADS	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**
Zone Conditioning	-	1540	365	-	464	0
Plenum Load	-	0	0	-	0	0
Return Fan Load	77 CFM	0	-	77 CFM	0	-
Ventilation Load	23 CFM	743	-365	23 CFM	1169	0
Supply Fan Load	77 CFM	115	-	77 CFM	-115	-
Zone Fan Coil Fans Load	-	0	-	-	0	-
>> Total System Loads	**-**	**2398**	**1**	**-**	**1518**	**0**
Central Cooling Coil	-	2405	0	-	0	0
Central Heating Coil	-	0	-	-	1523	-
>> Total Conditioning	**-**	**2405**	**0**	**-**	**1523**	**0**
Key:	**Positive values are cooling loads Negative values are heating loads**			**Positive values are heating loads Negative values are cooling loads**		

Table 2. Zone Heat Balance Loads

	DESIGN COOLING - JULY 16:00			DESIGN HEATING		
	OA DB / WB 106.2 F / 66.2 F			OA DB / WB 20.2 F / 16.3 F		
Zone Heat Balance Component	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**
Exterior Wall Convection	0 sqft	0	-	0 sqft	0	-
Roof Convection	0 sqft	0	-	0 sqft	0	-
Window Convection	0 sqft	0	-	0 sqft	0	-
Skylight Convection	0 sqft	0	-	0 sqft	0	-
Door Convection	0 sqft	0	-	0 sqft	0	-
Floor Convection	233 sqft	142	-	233 sqft	45	-
Interior Wall Convection	688 sqft	304	-	688 sqft	277	-
Ceiling Convection	233 sqft	316	-	233 sqft	114	-
Overhead Lighting Convection	96 W	145	-	0 W	0	-
Task Lighting Convection	0 W	0	-	0 W	0	-
Electric Equipment Convection	116 W	298	-	0 W	0	-
People Convection	2	137	381	0	0	0
Infiltration	0 CFM	0	0	0 CFM	0	0
Miscellaneous Equipment	-	0	0	-	0	0
Air Internal Energy Change	-	0		-	0	0
Safety Factor	0% / 0%	0	0	0%	0	0
>> Total Zone Loads	**-**	**1341**	**381**	**-**	**437**	**0**
Key:	**Positive values are cooling loads Negative values are heating loads**			**Positive values are heating loads Negative values are cooling loads**		

Note 1: Surface convection line items show the combined effects of conductive heat gain to the surface and radiative heat gains absorbed at the surface which are then convected to room air.

Note 2: Lighting, equipment, and people line items include only the direct convective heat gain from the heat source to the room air. The radiative portion of the heat gain is first absorbed by surfaces in the room and then later convected from the surface to the air. Therefore the effect of the radiative portion of the heat gain is found in the surface convection line items.

Note 3: Solar heat gain is absorbed by surfaces in the room, re-radiated to other surfaces, and finally convected from the surfaces to room air. Therefore, the effect of solar heat gain is found in the surface convection line items.

Air System Sizing Summary for HP-STAIRS1

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton) - Copy(converted)	06/11/2026
Prepared by: Shakespeare Engineering	11:07 AM

Air System Information

Air System Name	**HP-STAIRS1**		Number of zones	**1**	
Equipment Class	**SPLT AHU**		Floor Area	**992.6**	sqft
Air System Type	**SZCAV**		Location	**St George Municipal, UT, USA**	

Sizing Calculation Information

Calculation Months	**Jan to Dec**	Zone CFM Sizing	**Sum of space airflow rates**
Sizing Data	**Calculated**	Space CFM Sizing	**Individual peak space loads**

Central Cooling Coil Sizing Data

Total coil load	**1.0**	Tons	Peak coil load occurs at	**July 13:00**	
Total coil load	**11.7**	MBH	OA DB / WB	**102.9 / 65.2**	F
Sensible coil load	**11.7**	MBH	Entering DB / WB	**73.0 / 55.2**	F
Coil CFM at peak load	**512**	CFM	Leaving DB / WB	**49.7 / 45.5**	F
Sum of peak zone CFM	**512**	CFM	Resulting RH	**33**	%
Sensible heat ratio	**1.000**		Design supply temp.	**55.0**	F
CFM/Ton	**523.9**		Zone T-stat Check	**1 of 1**	OK
sqft/Ton	**1016.4**		Max zone temperature deviation	**0.0**	F
BTU/(hr sqft)	**11.8**				
Water flow @ 10.0 F rise	**N/A**				

Central Heating Coil Sizing Data

Max coil load	**8.4**	MBH	Load occurs at	**Design Heating**	
Coil CFM at Design Heating	**512**	CFM	BTU/(hr sqft)	**8.5**	
Max coil CFM	**512**	CFM	Ent. DB / Lvg DB	**72.0 / 88.9**	F
Water flow @ 20.0 F drop	**N/A**				

Supply Fan Sizing Data

Design CFM	**512**	CFM	Fan motor BHP	**0.28**	BHP
Design CFM/sqft	**0.52**	CFM/sqft	Fan motor kW	**0.22**	kW
			Fan total static	**2.00**	in wg

Outdoor Ventilation Air Data

Design airflow CFM	**0**	CFM	CFM/person	**0.00**	CFM/person
CFM/sqft	**0.00**	CFM/sqft			

Air System Information

Air System Name	**HP-STAIRS1**	Number of zones	**1**
Equipment Class	**SPLT AHU**	Floor Area	**992.6** sqft
Air System Type	**SZCAV**	Location	**St George Municipal, UT, USA**

Sizing Calculation Information

Calculation Months	**Jan to Dec**	Zone CFM Sizing	**Sum of space airflow rates**
Sizing Data	**Calculated**	Space CFM Sizing	**Individual peak space loads**

Zone Terminal Sizing Data

Zone Name	Design Supply Airflow (CFM)	Minimum Supply Airflow (CFM)	Zone CFM/sqft	Reheat Coil Capacity (MBH)	Reheat Coil Water gpm @ 20.0 F	Zone Htg Unit Coil Capacity (MBH)	Zone Htg Unit Water gpm @ 20.0 F	Mixing Box Fan Airflow (CFM)
HP-STAIRS1	512	512	0.52	0.0	-	0.0	-	0

Zone Peak Sensible Loads

Zone Name	Zone Cooling Sensible (MBH)	Time of Peak Sensible Cooling Load	Zone Heating Load (MBH)	Zone Floor Area (sqft)
HP-STAIRS1	8.3	July 12:00	7.3	992.6

Space Loads and Airflows

Zone Name / Space Name	Cooling Sensible (MBH)	Time of Peak Sensible Load	Air Flow (CFM)	Heating Load (MBH)	Floor Area (sqft)	Space CFM/sqft
HP-STAIRS1						
109-STAIRS 1	1.9	July 12:00	139	2.4	196.0	0.71
200-STAIRS 1	1.6	July 12:00	90	1.1	196.0	0.46
300-STAIRS 1	1.6	July 12:00	90	1.1	201.7	0.44
400-STAIRS 1	1.6	July 12:00	90	1.1	203.1	0.44
500-STAIRS 1	1.8	July 13:00	102	1.5	196.0	0.52

Ventilation Sizing Summary for HP-STAIRS1

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton) - Copy(converted) 06/11/2026

Prepared by: Shakespeare Engineering 11:07 AM

1. Summary

Ventilation Sizing Method .... **Sum of Space OA Airflows**

Design Ventilation Airflow Rate .... **0** CFM

2. Space Ventilation Analysis

Zone Name / Space Name	Floor Area (sqft)	Maximum Occupants	Maximum Supply Air (CFM)	Required Outdoor Air (CFM/person)	Required Outdoor Air (CFM/sqft)	Required Outdoor Air (CFM)	Required Outdoor Air (% of supply)	Required Outdoor Air (ACH)	Uncorrected Outdoor Air (CFM)	Direct Exhaust Air (CFM)
HP-STAIRS1										
109-STAIRS 1	196.0	0.0	139.1	0.00	0.00	0.0	0.0	0.00	0.0	0.0
200-STAIRS 1	196.0	0.0	90.4	0.00	0.00	0.0	0.0	0.00	0.0	0.0
300-STAIRS 1	201.7	0.0	89.7	0.00	0.00	0.0	0.0	0.00	0.0	0.0
400-STAIRS 1	203.1	0.0	90.1	0.00	0.00	0.0	0.0	0.00	0.0	0.0
500-STAIRS 1	196.0	0.0	102.4	0.00	0.00	0.0	0.0	0.00	0.0	0.0
Totals	**992.6**	**0.0**	**511.7**						**0.0**	

Air System Heat Balance Summary for HP-STAIRS1

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton) - Copy(converted) 06/11/2026
Prepared by: Shakespeare Engineering 11:07 AM

Table 1. System Loads

	DESIGN COOLING - JULY 13:00			DESIGN HEATING		
	OA DB / WB 102.9 F / 65.2 F			OA DB / WB 20.2 F / 16.3 F		
COMPONENT LOADS	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**
Zone Conditioning	-	10960	0	-	9173	0
Plenum Load	-	0	0	-	0	0
Return Fan Load	512 CFM	0	-	512 CFM	0	-
Ventilation Load	0 CFM	0	0	0 CFM	0	0
Supply Fan Load	512 CFM	759	-	512 CFM	-759	-
Zone Fan Coil Fans Load	-	0	-	-	0	-
>> Total System Loads	**-**	**11719**	**0**	**-**	**8415**	**0**
Central Cooling Coil	-	11719	0	-	0	0
Central Heating Coil	-	0	-	-	8415	-
>> Total Conditioning	**-**	**11719**	**0**	**-**	**8415**	**0**
Key:	**Positive values are cooling loads Negative values are heating loads**			**Positive values are heating loads Negative values are cooling loads**		

Table 2. Zone Heat Balance Loads

	DESIGN COOLING - JULY 13:00			DESIGN HEATING		
	OA DB / WB 102.9 F / 65.2 F			OA DB / WB 20.2 F / 16.3 F		
Zone Heat Balance Component	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**
Exterior Wall Convection	1263 sqft	3618	-	1263 sqft	3274	-
Roof Convection	0 sqft	0	-	0 sqft	0	-
Window Convection	0 sqft	0	-	0 sqft	0	-
Skylight Convection	0 sqft	0	-	0 sqft	0	-
Door Convection	20 sqft	92	-	20 sqft	128	-
Floor Convection	993 sqft	1207	-	993 sqft	1729	-
Interior Wall Convection	1296 sqft	972	-	1296 sqft	709	-
Ceiling Convection	995 sqft	1887	-	995 sqft	1440	-
Overhead Lighting Convection	328 W	497	-	0 W	0	-
Task Lighting Convection	0 W	0	-	0 W	0	-
Electric Equipment Convection	0 W	0	-	0 W	0	-
People Convection	0	0	0	0	0	0
Infiltration	0 CFM	0	0	0 CFM	0	0
Miscellaneous Equipment	-	0	0	-	0	0
Air Internal Energy Change	-	0		-	0	0
Safety Factor	0% / 0%	0	0	0%	0	0
>> Total Zone Loads	**-**	**8272**	**0**	**-**	**7281**	**0**
Key:	**Positive values are cooling loads Negative values are heating loads**			**Positive values are heating loads Negative values are cooling loads**		

Note 1: Surface convection line items show the combined effects of conductive heat gain to the surface and radiative heat gains absorbed at the surface which are then convected to room air.

Note 2: Lighting, equipment, and people line items include only the direct convective heat gain from the heat source to the room air. The radiative portion of the heat gain is first absorbed by surfaces in the room and then later convected from the surface to the air. Therefore the effect of the radiative portion of the heat gain is found in the surface convection line items.

Note 3: Solar heat gain is absorbed by surfaces in the room, re-radiated to other surfaces, and finally convected from the surfaces to room air. Therefore, the effect of solar heat gain is found in the surface convection line items.

Air System Sizing Summary for HP-STAIRS2

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton) - Copy(converted) 06/11/2026
Prepared by: Shakespeare Engineering 11:07 AM

Air System Information

Air System Name	**HP-STAIRS2**		Number of zones	**1**	
Equipment Class	**SPLT AHU**		Floor Area	**1000.7**	sqft
Air System Type	**SZCAV**		Location	**St George Municipal, UT, USA**	

Sizing Calculation Information

Calculation Months	**Jan to Dec**	Zone CFM Sizing	**Sum of space airflow rates**
Sizing Data	**Calculated**	Space CFM Sizing	**Individual peak space loads**

Central Cooling Coil Sizing Data

Total coil load	**1.2**	Tons	Peak coil load occurs at	**July 19:00**	
Total coil load	**14.3**	MBH	OA DB / WB	**100.1 / 64.3**	F
Sensible coil load	**14.3**	MBH	Entering DB / WB	**73.0 / 55.2**	F
Coil CFM at peak load	**607**	CFM	Leaving DB / WB	**49.1 / 45.2**	F
Sum of peak zone CFM	**607**	CFM	Resulting RH	**33**	%
Sensible heat ratio	**1.000**		Design supply temp.	**55.0**	F
CFM/Ton	**509.8**		Zone T-stat Check	**1 of 1**	OK
sqft/Ton	**840.6**		Max zone temperature deviation	**0.0**	F
BTU/(hr sqft)	**14.3**				
Water flow @ 10.0 F rise	**N/A**				

Central Heating Coil Sizing Data

Max coil load	**8.6**	MBH	Load occurs at	**Design Heating**	
Coil CFM at Design Heating	**607**	CFM	BTU/(hr sqft)	**8.5**	
Max coil CFM	**607**	CFM	Ent. DB / Lvg DB	**72.0 / 86.5**	F
Water flow @ 20.0 F drop	**N/A**				

Supply Fan Sizing Data

Design CFM	**607**	CFM	Fan motor BHP	**0.33**	BHP
Design CFM/sqft	**0.61**	CFM/sqft	Fan motor kW	**0.26**	kW
			Fan total static	**2.00**	in wg

Outdoor Ventilation Air Data

Design airflow CFM	**0**	CFM	CFM/person	**0.00**	CFM/person
CFM/sqft	**0.00**	CFM/sqft			

Air System Information

Air System Name	**HP-STAIRS2**	Number of zones	**1**
Equipment Class	**SPLT AHU**	Floor Area	**1000.7** sqft
Air System Type	**SZCAV**	Location	**St George Municipal, UT, USA**

Sizing Calculation Information

Calculation Months	**Jan to Dec**	Zone CFM Sizing	**Sum of space airflow rates**
Sizing Data	**Calculated**	Space CFM Sizing	**Individual peak space loads**

Zone Terminal Sizing Data

Zone Name	Design Supply Airflow (CFM)	Minimum Supply Airflow (CFM)	Zone CFM/sqft	Reheat Coil Capacity (MBH)	Reheat Coil Water gpm @ 20.0 F	Zone Htg Unit Coil Capacity (MBH)	Zone Htg Unit Water gpm @ 20.0 F	Mixing Box Fan Airflow (CFM)
HP-STAIRS2	607	607	0.61	0.0	-	0.0	-	0

Zone Peak Sensible Loads

Zone Name	Zone Cooling Sensible (MBH)	Time of Peak Sensible Cooling Load	Zone Heating Load (MBH)	Zone Floor Area (sqft)
HP-STAIRS2	10.6	August 18:00	7.4	1000.7

Space Loads and Airflows

Zone Name / Space Name	Cooling Sensible (MBH)	Time of Peak Sensible Load	Air Flow (CFM)	Heating Load (MBH)	Floor Area (sqft)	Space CFM/sqft
HP-STAIRS2						
143-STAIRS 2	2.5	August 18:00	141	2.5	201.3	0.70
210-STAIRS 2	2.0	August 19:00	112	1.1	202.7	0.55
310-STAIRS 2	2.0	August 19:00	112	1.1	198.6	0.56
410-STAIRS 2	1.9	August 19:00	110	1.1	199.1	0.55
510 STAIRS 2	2.3	August 18:00	132	1.5	199.1	0.66

1. Summary

Ventilation Sizing Method .... **Sum of Space OA Airflows**
Design Ventilation Airflow Rate .... **0** CFM

2. Space Ventilation Analysis

Zone Name / Space Name	Floor Area (sqft)	Maximum Occupants	Maximum Supply Air (CFM)	Required Outdoor Air (CFM/person)	Required Outdoor Air (CFM/sqft)	Required Outdoor Air (CFM)	Required Outdoor Air (% of supply)	Required Outdoor Air (ACH)	Uncorrected Outdoor Air (CFM)	Direct Exhaust Air (CFM)
HP-STAIRS2										
143-STAIRS 2	201.3	0.0	141.1	0.00	0.00	0.0	0.0	0.00	0.0	0.0
210-STAIRS 2	202.7	0.0	112.2	0.00	0.00	0.0	0.0	0.00	0.0	0.0
310-STAIRS 2	198.6	0.0	111.8	0.00	0.00	0.0	0.0	0.00	0.0	0.0
410-STAIRS 2	199.1	0.0	109.7	0.00	0.00	0.0	0.0	0.00	0.0	0.0
510 STAIRS 2	199.1	0.0	132.1	0.00	0.00	0.0	0.0	0.00	0.0	0.0
Totals	**1000.7**	**0.0**	**606.9**						**0.0**	

Table 1. System Loads

	DESIGN COOLING - JULY 19:00			DESIGN HEATING		
	OA DB / WB 100.1 F / 64.3 F			OA DB / WB 20.2 F / 16.3 F		
COMPONENT LOADS	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**
Zone Conditioning	-	13385	0	-	9451	0
Plenum Load	-	0	0	-	0	0
Return Fan Load	607 CFM	0	-	607 CFM	0	-
Ventilation Load	0 CFM	0	0	0 CFM	0	0
Supply Fan Load	607 CFM	900	-	607 CFM	-900	-
Zone Fan Coil Fans Load	-	0	-	-	0	-
>> Total System Loads	**-**	**14285**	**0**	**-**	**8551**	**0**
Central Cooling Coil	-	14285	0	-	0	0
Central Heating Coil	-	0	-	-	8551	-
>> Total Conditioning	**-**	**14285**	**0**	**-**	**8551**	**0**
Key:	**Positive values are cooling loads Negative values are heating loads**			**Positive values are heating loads Negative values are cooling loads**		

Table 2. Zone Heat Balance Loads

	DESIGN COOLING - JULY 19:00			DESIGN HEATING		
	OA DB / WB 100.1 F / 64.3 F			OA DB / WB 20.2 F / 16.3 F		
Zone Heat Balance Component	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**
Exterior Wall Convection	1269 sqft	4916	-	1269 sqft	3303	-
Roof Convection	0 sqft	0	-	0 sqft	0	-
Window Convection	0 sqft	0	-	0 sqft	0	-
Skylight Convection	0 sqft	0	-	0 sqft	0	-
Door Convection	20 sqft	161	-	20 sqft	130	-
Floor Convection	1001 sqft	1354	-	1001 sqft	1757	-
Interior Wall Convection	1293 sqft	1168	-	1293 sqft	726	-
Ceiling Convection	1002 sqft	2242	-	1002 sqft	1463	-
Overhead Lighting Convection	331 W	501	-	0 W	0	-
Task Lighting Convection	0 W	0	-	0 W	0	-
Electric Equipment Convection	0 W	0	-	0 W	0	-
People Convection	0	0	0	0	0	0
Infiltration	0 CFM	0	0	0 CFM	0	0
Miscellaneous Equipment	-	0	0	-	0	0
Air Internal Energy Change	-	0		-	0	0
Safety Factor	0% / 0%	0	0	0%	0	0
>> Total Zone Loads	**-**	**10342**	**0**	**-**	**7379**	**0**
Key:	**Positive values are cooling loads Negative values are heating loads**			**Positive values are heating loads Negative values are cooling loads**		

Note 1: Surface convection line items show the combined effects of conductive heat gain to the surface and radiative heat gains absorbed at the surface which are then convected to room air.

Note 2: Lighting, equipment, and people line items include only the direct convective heat gain from the heat source to the room air. The radiative portion of the heat gain is first absorbed by surfaces in the room and then later convected from the surface to the air. Therefore the effect of the radiative portion of the heat gain is found in the surface convection line items.

Note 3: Solar heat gain is absorbed by surfaces in the room, re-radiated to other surfaces, and finally convected from the surfaces to room air. Therefore, the effect of solar heat gain is found in the surface convection line items.

Air System Sizing Summary for UH-FIRE RISER

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton) - Copy(converted) 06/11/2026
Prepared by: Shakespeare Engineering 11:07 AM

Air System Information

Air System Name	**UH-FIRE RISER**		Number of zones	**1**	
Equipment Class	**UNDEF**		Floor Area	**35.0**	sqft
Air System Type	**SZCAV**		Location	**St George Municipal, UT, USA**	

Sizing Calculation Information

Calculation Months	**Jan to Dec**	Zone CFM Sizing	**Sum of space airflow rates**
Sizing Data	**Calculated**	Space CFM Sizing	**Individual peak space loads**

Central Heating Coil Sizing Data

Max coil load	**0.8**	MBH	Load occurs at	**Design Heating**	
Coil CFM at Design Heating	**48**	CFM	BTU/(hr sqft)	**22.7**	
Max coil CFM	**48**	CFM	Ent. DB / Lvg DB	**69.7 / 86.6**	F
Water flow @ 20.0 F drop	**0.08**	gpm			

Supply Fan Sizing Data

Design CFM	**48**	CFM	Fan motor BHP	**0.03**	BHP
Design CFM/sqft	**1.38**	CFM/sqft	Fan motor kW	**0.02**	kW
			Fan total static	**2.00**	in wg

Outdoor Ventilation Air Data

Design airflow CFM	**2**	CFM	CFM/person	**0.00**	CFM/person
CFM/sqft	**0.06**	CFM/sqft			

Zone Sizing Summary for UH-FIRE RISER

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton) - Copy(converted) 06/11/2026
Prepared by: Shakespeare Engineering 11:07 AM

Air System Information

Air System Name .... **UH-FIRE RISER**
Equipment Class .... **UNDEF**
Air System Type .... **SZCAV**

Number of zones **1**
Floor Area .... **35.0** sqft
Location .... **St George Municipal, UT, USA**

Sizing Calculation Information

Calculation Months .... **Jan to Dec**
Sizing Data .... **Calculated**

Zone CFM Sizing .... **Sum of space airflow rates**
Space CFM Sizing .... **Individual peak space loads**

Zone Terminal Sizing Data

Zone Name	Design Supply Airflow (CFM)	Minimum Supply Airflow (CFM)	Zone CFM/sqft	Reheat Coil Capacity (MBH)	Reheat Coil Water gpm @ 20.0 F	Zone Htg Unit Coil Capacity (MBH)	Zone Htg Unit Water gpm @ 20.0 F	Mixing Box Fan Airflow (CFM)
UH-FIRE RISER	48	48	1.38	0.0	0.00	0.0	0.00	0

Zone Peak Sensible Loads

Zone Name	Zone Cooling Sensible (MBH)	Time of Peak Sensible Cooling Load	Zone Heating Load (MBH)	Zone Floor Area (sqft)
UH-FIRE RISER	0.0	March 12:00	0.8	35.0

Space Loads and Airflows

Zone Name / Space Name	Cooling Sensible (MBH)	Time of Peak Sensible Load	Air Flow (CFM)	Heating Load (MBH)	Floor Area (sqft)	Space CFM/sqft
UH-FIRE RISER						
136-FIRE RISER	0.0	March 12:00	48	0.8	35.0	1.38

Ventilation Sizing Summary for UH-FIRE RISER
(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton) - Copy(converted)	06/11/2026
Prepared by: Shakespeare Engineering	11:07 AM

1. Summary

Ventilation Sizing Method .... **Sum of Space OA Airflows**
Design Ventilation Airflow Rate .... **2** CFM

2. Space Ventilation Analysis

Zone Name / Space Name	Floor Area (sqft)	Maximum Occupants	Maximum Supply Air (CFM)	Required Outdoor Air (CFM/person)	Required Outdoor Air (CFM/sqft)	Required Outdoor Air (CFM)	Required Outdoor Air (% of supply)	Required Outdoor Air (ACH)	Uncorrected Outdoor Air (CFM)	Direct Exhaust Air (CFM)
UH-FIRE RISER										
136-FIRE RISER	35.0	0.0	48.3	5.00	0.06	0.0	0.0	0.00	2.1	0.0
Totals	**35.0**	**0.0**	**48.3**						**2.1**	

Air System Heat Balance Summary for UH-FIRE RISER
(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton) - Copy(converted)
Prepared by: Shakespeare Engineering

Table 1. System Loads

	DESIGN COOLING			DESIGN HEATING		
	NO COOLING DATA			OA DB / WB 20.2 F / 16.3 F		
COMPONENT LOADS	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**
Zone Conditioning	-	-	-	-	760	0
Plenum Load	-	-	-	-	0	0
Return Fan Load	-	-	-	48 CFM	0	-
Ventilation Load	-	-	-	2 CFM	106	0
Supply Fan Load	-	-	-	48 CFM	-72	-
Zone Fan Coil Fans Load	-	-	-	-	0	-
>> Total System Loads	**-**	**-**	**-**	**-**	**794**	**0**
Central Heating Coil	-	-	-	-	794	-
>> Total Conditioning	**-**	**-**	**-**	**-**	**794**	**0**
Key:	**Positive values are cooling loads Negative values are heating loads**			**Positive values are heating loads Negative values are cooling loads**		

Table 2. Zone Heat Balance Loads

	DESIGN COOLING			DESIGN HEATING		
	NO COOLING DATA			OA DB / WB 20.2 F / 16.3 F		
Zone Heat Balance Component	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**
Exterior Wall Convection	54 sqft	-	-	54 sqft	172	-
Roof Convection	0 sqft	-	-	0 sqft	0	-
Window Convection	0 sqft	-	-	0 sqft	0	-
Skylight Convection	0 sqft	-	-	0 sqft	0	-
Door Convection	20 sqft	-	-	20 sqft	152	-
Floor Convection	35 sqft	-	-	35 sqft	224	-
Interior Wall Convection	169 sqft	-	-	169 sqft	237	-
Ceiling Convection	35 sqft	-	-	35 sqft	59	-
Overhead Lighting Convection	-	-	-	0 W	0	-
Task Lighting Convection	-	-	-	0 W	0	-
Electric Equipment Convection	-	-	-	0 W	0	-
People Convection	-	-	-	0	0	0
Infiltration	-	-	-	0 CFM	0	0
Miscellaneous Equipment	-	-	-	-	0	0
Air Internal Energy Change	-	-		-	0	0
Safety Factor	0% / 0%	-	-	0%	0	0
>> Total Zone Loads	**-**	**-**	**-**	**-**	**844**	**0**
Key:	**Positive values are cooling loads Negative values are heating loads**			**Positive values are heating loads Negative values are cooling loads**		

Note 1: Surface convection line items show the combined effects of conductive heat gain to the surface and radiative heat gains absorbed at the surface which are then convected to room air.

Note 2: Lighting, equipment, and people line items include only the direct convective heat gain from the heat source to the room air. The radiative portion of the heat gain is first absorbed by surfaces in the room and then later convected from the surface to the air. Therefore the effect of the radiative portion of the heat gain is found in the surface convection line items.

Note 3: Solar heat gain is absorbed by surfaces in the room, re-radiated to other surfaces, and finally convected from the surfaces to room air. Therefore, the effect of solar heat gain is found in the surface convection line items.

Air System Sizing Summary for UH-POOL EQUIP STORAGE

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton) - Copy(converted) 06/11/2026
Prepared by: Shakespeare Engineering 11:07 AM

Air System Information

Air System Name	**UH-POOL EQUIP STORAGE**		Number of zones	**1**	
Equipment Class	**UNDEF**		Floor Area	**102.9**	sqft
Air System Type	**SZCAV**		Location	**St George Municipal, UT, USA**	

Sizing Calculation Information

Calculation Months	**Jan to Dec**	Zone CFM Sizing	**Sum of space airflow rates**
Sizing Data	**Calculated**	Space CFM Sizing	**Individual peak space loads**

Central Heating Coil Sizing Data

Max coil load	**1.5**	MBH	Load occurs at	**Design Heating**	
Coil CFM at Design Heating	**75**	CFM	BTU/(hr sqft)	**14.7**	
Max coil CFM	**75**	CFM	Ent. DB / Lvg DB	**67.7 / 88.4**	F
Water flow @ 20.0 F drop	**0.15**	gpm			

Supply Fan Sizing Data

Design CFM	**75**	CFM	Fan motor BHP	**0.04**	BHP
Design CFM/sqft	**0.73**	CFM/sqft	Fan motor kW	**0.03**	kW
			Fan total static	**2.00**	in wg

Outdoor Ventilation Air Data

Design airflow CFM	**6**	CFM	CFM/person	**0.00**	CFM/person
CFM/sqft	**0.06**	CFM/sqft			

Air System Information

Air System Name .... **UH-POOL EQUIP STORAGE**
Equipment Class .... **UNDEF**
Air System Type .... **SZCAV**

Number of zones **1**
Floor Area .... **102.9** sqft
Location .... **St George Municipal, UT, USA**

Sizing Calculation Information

Calculation Months .... **Jan to Dec**
Sizing Data .... **Calculated**

Zone CFM Sizing .... **Sum of space airflow rates**
Space CFM Sizing .... **Individual peak space loads**

Zone Terminal Sizing Data

Zone Name	Design Supply Airflow (CFM)	Minimum Supply Airflow (CFM)	Zone CFM/sqft	Reheat Coil Capacity (MBH)	Reheat Coil Water gpm @ 20.0 F	Zone Htg Unit Coil Capacity (MBH)	Zone Htg Unit Water gpm @ 20.0 F	Mixing Box Fan Airflow (CFM)
UH-POOL STORAGE	75	75	0.73	0.0	0.00	0.0	0.00	0

Zone Peak Sensible Loads

Zone Name	Zone Cooling Sensible (MBH)	Time of Peak Sensible Cooling Load	Zone Heating Load (MBH)	Zone Floor Area (sqft)
UH-POOL STORAGE	0.0	November 23:00	1.3	102.9

Space Loads and Airflows

Zone Name / Space Name	Cooling Sensible (MBH)	Time of Peak Sensible Load	Air Flow (CFM)	Heating Load (MBH)	Floor Area (sqft)	Space CFM/sqft
UH-POOL STORAGE						
POOL STORAGE	0.0	November 23:00	75	1.3	102.9	0.73

1. Summary

Ventilation Sizing Method .... **Sum of Space OA Airflows**
Design Ventilation Airflow Rate .... **6** CFM

2. Space Ventilation Analysis

Zone Name / Space Name	Floor Area (sqft)	Maximum Occupants	Maximum Supply Air (CFM)	Required Outdoor Air (CFM/person)	Required Outdoor Air (CFM/sqft)	Required Outdoor Air (CFM)	Required Outdoor Air (% of supply)	Required Outdoor Air (ACH)	Uncorrected Outdoor Air (CFM)	Direct Exhaust Air (CFM)
UH-POOL STORAGE										
POOL STORAGE	102.9	0.0	74.9	5.00	0.06	0.0	0.0	0.00	6.2	0.0
Totals	**102.9**	**0.0**	**74.9**						**6.2**	

Table 1. System Loads

	DESIGN COOLING			DESIGN HEATING		
	NO COOLING DATA			OA DB / WB 20.2 F / 16.3 F		
COMPONENT LOADS	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**
Zone Conditioning	-	-	-	-	1308	0
Plenum Load	-	-	-	-	0	0
Return Fan Load	-	-	-	75 CFM	0	-
Ventilation Load	-	-	-	6 CFM	310	0
Supply Fan Load	-	-	-	75 CFM	-111	-
Zone Fan Coil Fans Load	-	-	-	-	0	-
>> Total System Loads	**-**	**-**	**-**	**-**	**1508**	**0**
Central Heating Coil	-	-	-	-	1509	-
>> Total Conditioning	**-**	**-**	**-**	**-**	**1509**	**0**
Key:	**Positive values are cooling loads Negative values are heating loads**			**Positive values are heating loads Negative values are cooling loads**		

Table 2. Zone Heat Balance Loads

	DESIGN COOLING			DESIGN HEATING		
	NO COOLING DATA			OA DB / WB 20.2 F / 16.3 F		
Zone Heat Balance Component	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**
Exterior Wall Convection	60 sqft	-	-	60 sqft	168	-
Roof Convection	0 sqft	-	-	0 sqft	0	-
Window Convection	0 sqft	-	-	0 sqft	0	-
Skylight Convection	0 sqft	-	-	0 sqft	0	-
Door Convection	20 sqft	-	-	20 sqft	128	-
Floor Convection	103 sqft	-	-	103 sqft	390	-
Interior Wall Convection	336 sqft	-	-	336 sqft	472	-
Ceiling Convection	103 sqft	-	-	103 sqft	151	-
Overhead Lighting Convection	-	-	-	0 W	0	-
Task Lighting Convection	-	-	-	0 W	0	-
Electric Equipment Convection	-	-	-	0 W	0	-
People Convection	-	-	-	0	0	0
Infiltration	-	-	-	0 CFM	0	0
Miscellaneous Equipment	-	-	-	-	0	0
Air Internal Energy Change	-	-		-	0	0
Safety Factor	0% / 0%	-	-	0%	0	0
>> Total Zone Loads	**-**	**-**	**-**	**-**	**1309**	**0**
Key:	**Positive values are cooling loads Negative values are heating loads**			**Positive values are heating loads Negative values are cooling loads**		

Note 1: Surface convection line items show the combined effects of conductive heat gain to the surface and radiative heat gains absorbed at the surface which are then convected to room air.

Note 2: Lighting, equipment, and people line items include only the direct convective heat gain from the heat source to the room air. The radiative portion of the heat gain is first absorbed by surfaces in the room and then later convected from the surface to the air. Therefore the effect of the radiative portion of the heat gain is found in the surface convection line items.

Note 3: Solar heat gain is absorbed by surfaces in the room, re-radiated to other surfaces, and finally convected from the surfaces to room air. Therefore, the effect of solar heat gain is found in the surface convection line items.

Air System Sizing Summary for UH-VESTIBULE

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton) - Copy(converted)	06/11/2026
Prepared by: Shakespeare Engineering	11:07 AM

Air System Information

Air System Name	**UH-VESTIBULE**		Number of zones	**1**	
Equipment Class	**UNDEF**		Floor Area	**155.1**	sqft
Air System Type	**SZCAV**		Location	**St George Municipal, UT, USA**	

Sizing Calculation Information

Calculation Months	**Jan to Dec**	Zone CFM Sizing	**Sum of space airflow rates**
Sizing Data	**Calculated**	Space CFM Sizing	**Individual peak space loads**

Central Heating Coil Sizing Data

Max coil load	**3.3**	MBH	Load occurs at	**Design Heating**	
Coil CFM at Design Heating	**170**	CFM	BTU/(hr sqft)	**21.5**	
Max coil CFM	**170**	CFM	Ent. DB / Lvg DB	**69.2 / 89.3**	F
Water flow @ 20.0 F drop	**0.33**	gpm			

Supply Fan Sizing Data

Design CFM	**170**	CFM	Fan motor BHP	**0.09**	BHP
Design CFM/sqft	**1.10**	CFM/sqft	Fan motor kW	**0.07**	kW
			Fan total static	**2.00**	in wg

Outdoor Ventilation Air Data

Design airflow CFM	**9**	CFM	CFM/person	**0.00**	CFM/person
CFM/sqft	**0.06**	CFM/sqft			

Air System Information

Air System Name .... **UH-VESTIBULE**
Equipment Class .... **UNDEF**
Air System Type .... **SZCAV**

Number of zones **1**
Floor Area .... **155.1** sqft
Location .... **St George Municipal, UT, USA**

Sizing Calculation Information

Calculation Months .... **Jan to Dec**
Sizing Data .... **Calculated**

Zone CFM Sizing .... **Sum of space airflow rates**
Space CFM Sizing .... **Individual peak space loads**

Zone Terminal Sizing Data

Zone Name	Design Supply Airflow (CFM)	Minimum Supply Airflow (CFM)	Zone CFM/sqft	Reheat Coil Capacity (MBH)	Reheat Coil Water gpm @ 20.0 F	Zone Htg Unit Coil Capacity (MBH)	Zone Htg Unit Water gpm @ 20.0 F	Mixing Box Fan Airflow (CFM)
UH-01-VESTIBULE	170	170	1.10	0.0	0.00	0.0	0.00	0

Zone Peak Sensible Loads

Zone Name	Zone Cooling Sensible (MBH)	Time of Peak Sensible Cooling Load	Zone Heating Load (MBH)	Zone Floor Area (sqft)
UH-01-VESTIBULE	0.0	May 6:00	3.0	155.1

Space Loads and Airflows

Zone Name / Space Name	Cooling Sensible (MBH)	Time of Peak Sensible Load	Air Flow (CFM)	Heating Load (MBH)	Floor Area (sqft)	Space CFM/sqft
UH-01-VESTIBULE						
100-VESTIBULE	0.0	May 6:00	170	3.0	155.1	1.10

Ventilation Sizing Summary for UH-VESTIBULE

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton) - Copy(converted) 06/11/2026

Prepared by: Shakespeare Engineering 11:07 AM

1. Summary

Ventilation Sizing Method .... **Sum of Space OA Airflows**

Design Ventilation Airflow Rate .... **9** CFM

2. Space Ventilation Analysis

Zone Name / Space Name	Floor Area (sqft)	Maximum Occupants	Maximum Supply Air (CFM)	Required Outdoor Air (CFM/person)	Required Outdoor Air (CFM/sqft)	Required Outdoor Air (CFM)	Required Outdoor Air (% of supply)	Required Outdoor Air (ACH)	Uncorrected Outdoor Air (CFM)	Direct Exhaust Air (CFM)
UH-01-VESTIBULE										
100-VESTIBULE	155.1	0.0	169.9	0.00	0.06	0.0	0.0	0.00	9.3	0.0
Totals	**155.1**	**0.0**	**169.9**						**9.3**	

Air System Heat Balance Summary for UH-VESTIBULE

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton) - Copy(converted) 06/11/2026
Prepared by: Shakespeare Engineering 11:07 AM

Table 1. System Loads

	DESIGN COOLING			DESIGN HEATING		
	NO COOLING DATA			OA DB / WB 20.2 F / 16.3 F		
COMPONENT LOADS	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**
Zone Conditioning	-	-	-	-	3110	0
Plenum Load	-	-	-	-	0	0
Return Fan Load	-	-	-	170 CFM	0	-
Ventilation Load	-	-	-	9 CFM	468	0
Supply Fan Load	-	-	-	170 CFM	-252	-
Zone Fan Coil Fans Load	-	-	-	-	0	-
>> Total System Loads	**-**	**-**	**-**	**-**	**3326**	**0**
Central Heating Coil	-	-	-	-	3328	-
>> Total Conditioning	**-**	**-**	**-**	**-**	**3328**	**0**
Key:	**Positive values are cooling loads Negative values are heating loads**			**Positive values are heating loads Negative values are cooling loads**		

Table 2. Zone Heat Balance Loads

	DESIGN COOLING			DESIGN HEATING		
	NO COOLING DATA			OA DB / WB 20.2 F / 16.3 F		
Zone Heat Balance Component	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**
Exterior Wall Convection	217 sqft	-	-	217 sqft	687	-
Roof Convection	0 sqft	-	-	0 sqft	0	-
Window Convection	87 sqft	-	-	87 sqft	943	-
Skylight Convection	0 sqft	-	-	0 sqft	0	-
Door Convection	0 sqft	-	-	0 sqft	0	-
Floor Convection	155 sqft	-	-	155 sqft	961	-
Interior Wall Convection	164 sqft	-	-	164 sqft	123	-
Ceiling Convection	155 sqft	-	-	155 sqft	253	-
Overhead Lighting Convection	-	-	-	0 W	0	-
Task Lighting Convection	-	-	-	0 W	0	-
Electric Equipment Convection	-	-	-	0 W	0	-
People Convection	-	-	-	0	0	0
Infiltration	-	-	-	0 CFM	0	0
Miscellaneous Equipment	-	-	-	-	0	0
Air Internal Energy Change	-	-		-	0	0
Safety Factor	0% / 0%	-	-	0%	0	0
>> Total Zone Loads	**-**	**-**	**-**	**-**	**2967**	**0**
Key:	**Positive values are cooling loads Negative values are heating loads**			**Positive values are heating loads Negative values are cooling loads**		

Note 1: Surface convection line items show the combined effects of conductive heat gain to the surface and radiative heat gains absorbed at the surface which are then convected to room air.

Note 2: Lighting, equipment, and people line items include only the direct convective heat gain from the heat source to the room air. The radiative portion of the heat gain is first absorbed by surfaces in the room and then later convected from the surface to the air. Therefore the effect of the radiative portion of the heat gain is found in the surface convection line items.

Note 3: Solar heat gain is absorbed by surfaces in the room, re-radiated to other surfaces, and finally convected from the surfaces to room air. Therefore, the effect of solar heat gain is found in the surface convection line items.

Air System Information

Air System Name	**UNIT KING**		Number of zones	**1**	
Equipment Class	**TERM**		Floor Area	**353.3**	sqft
Air System Type	**PKG-FC**		Location	**St George Municipal, UT, USA**	

Sizing Calculation Information

Calculation Months	**Jan to Dec**	Zone CFM Sizing	**Sum of space airflow rates**
Sizing Data	**Calculated**	Space CFM Sizing	**Individual peak space loads**

Cooling Coil Sizing Data

Total coil load	**0.1**	Tons	Load occurs at	**July 15:00**	
Total coil load	**1.2**	MBH	OA DB / WB	**106.2 / 66.2**	F
Total coil load	**374.2**	CFM/Ton	Entering DB / WB	**106.1 / 66.2**	F
Sensible coil load	**1.2**	MBH	Leaving DB / WB	**73.5 / 55.4**	F
Coil CFM at July 15:00	**39**	CFM			
Max coil CFM	**39**	CFM			
Sensible heat ratio	**1.000**				
Water flow @ 10.0 F rise	**N/A**				

Heating Coil Sizing Data

Max coil load	**1.8**	MBH	Load occurs at	**Design Heating**	
Coil CFM at Design Heating	**39**	CFM	Ent. DB / Lvg DB	**20.0 / 68.5**	F
Max coil CFM	**39**	CFM			
Water flow @ 20.0 F drop	**N/A**				

Ventilation Fan Sizing Data

Design CFM	**39**	CFM	Fan motor BHP	**0.02**	BHP
Design CFM/sqft	**0.11**	CFM/sqft	Fan motor kW	**0.02**	kW
			Fan total static	**2.00**	in wg

Outdoor Ventilation Air Data

Design airflow CFM	**39**	CFM	CFM/person	**11.00**	CFM/person
CFM/sqft	**0.11**	CFM/sqft			

Air System Information

Air System Name: **UNIT KING**
Equipment Class: **TERM**
Air System Type: **PKG-FC**

Number of zones: **1**
Floor Area: **353.3** sqft
Location: **St George Municipal, UT, USA**

Sizing Calculation Information

Calculation Months: **Jan to Dec**
Sizing Data: **Calculated**

Zone CFM Sizing: **Sum of space airflow rates**
Space CFM Sizing: **Individual peak space loads**

Terminal Unit Sizing Data - Cooling

Zone Name	Total Coil Load (MBH)	Sens Coil Load (MBH)	Coil Entering DB / WB (F)	Coil Leaving DB / WB (F)	Water Flow @ 10.0 F (gpm)	Time of Peak Coil Load	Zone CFM/sqft
UNIT KING	6.7	6.4	73.4 / 56.6	56.7 / 49.7	-	July 19:00	1.10

Terminal Unit Sizing Data - Heating, Fan, Ventilation

Zone Name	Heating Coil Load (MBH)	Heating Coil Ent/Lvg DB (F)	Htg Coil Water Flow @20.0 F (gpm)	Fan Design Airflow (CFM)	Fan Motor (BHP)	Fan Motor (kW)	OA Vent Design Airflow (CFM)
UNIT KING	3.3	72.0 / 80.6	-	390	0.029	0.023	39

Zone Peak Sensible Loads

Zone Name	Zone Cooling Sensible (MBH)	Time of Peak Sensible Cooling Load	Zone Heating Load (MBH)	Zone Floor Area (sqft)
UNIT KING	5.7	June 19:00	2.6	353.3

Space Loads and Airflows

Zone Name / Space Name	Cooling Sensible (MBH)	Time of Peak Sensible Load	Air Flow (CFM)	Heating Load (MBH)	Floor Area (sqft)	Space CFM/sqft
UNIT KING						
UNIT 533-KING	5.7	June 19:00	390	2.6	353.3	1.10

Ventilation Sizing Summary for UNIT KING
(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton) - Copy(converted) 06/11/2026
Prepared by: Shakespeare Engineering 11:07 AM

1. Summary

Ventilation Sizing Method .... **Sum of Space OA Airflows**
Design Ventilation Airflow Rate .... **39** CFM

2. Space Ventilation Analysis

2.1 Zone: UNIT KING

Zone Name / Space Name	Floor Area (sqft)	Maximum Occupants	Maximum Supply Air (CFM)	Required Outdoor Air (CFM/person)	Required Outdoor Air (CFM/sqft)	Required Outdoor Air (CFM)	Required Outdoor Air (% of supply)	Required Outdoor Air (ACH)	Uncorrected Outdoor Air (CFM)	Direct Exhaust Air (CFM)
UNIT KING										
UNIT 533-KING	353.3	3.5	390.1	5.00	0.06	0.0	0.0	0.00	38.9	0.0
Totals	**353.3**	**3.5**	**390.1**						**38.9**	**0.0**

Table 1. System Loads

	DESIGN COOLING - JULY 19:00			DESIGN HEATING		
	OA DB / WB 100.1 F / 64.3 F			OA DB / WB 20.2 F / 16.3 F		
COMPONENT LOADS	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**
Zone Conditioning	-	6263	426	-	3255	0
Plenum Load	-	0	0	-	0	0
Exhaust Fan Load	39 CFM	0	-	39 CFM	0	-
Ventilation Load	39 CFM	1006	-138	39 CFM	1953	0
Ventilation Fan Load	39 CFM	58	-	39 CFM	-58	-
Zone Fan Coil Fans Load	-	78	-	-	-78	-
>> Total System Loads	**-**	**7404**	**288**	**-**	**5073**	**0**
Cooling Coil	-	1001	0	-	0	0
Heating Coil	-	0	-	-	1828	-
Terminal Unit Cooling	-	6414	324	-	0	0
Terminal Unit Heating	-	0	-	-	3253	-
>> Total Conditioning	**-**	**7415**	**324**	**-**	**5081**	**0**
Key:	**Positive values are cooling loads Negative values are heating loads**			**Positive values are heating loads Negative values are cooling loads**		

Table 2. Zone Heat Balance Loads

	DESIGN COOLING - JULY 19:00			DESIGN HEATING		
	OA DB / WB 100.1 F / 64.3 F			OA DB / WB 20.2 F / 16.3 F		
Zone Heat Balance Component	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**
Exterior Wall Convection	294 sqft	1439	-	294 sqft	792	-
Roof Convection	0 sqft	0	-	0 sqft	0	-
Window Convection	36 sqft	304	-	36 sqft	319	-
Skylight Convection	0 sqft	0	-	0 sqft	0	-
Door Convection	0 sqft	0	-	0 sqft	0	-
Floor Convection	353 sqft	599	-	353 sqft	355	-
Interior Wall Convection	330 sqft	483	-	330 sqft	230	-
Ceiling Convection	353 sqft	1516	-	353 sqft	942	-
Overhead Lighting Convection	98 W	148	-	0 W	0	-
Task Lighting Convection	0 W	0	-	0 W	0	-
Electric Equipment Convection	353 W	904	-	0 W	0	-
People Convection	4	244	424	0	0	0
Infiltration	0 CFM	0	0	0 CFM	0	0
Miscellaneous Equipment	-	0	0	-	0	0
Air Internal Energy Change	-	0		-	0	0
Safety Factor	0% / 0%	0	0	0%	0	0
>> Total Zone Loads	**-**	**5637**	**424**	**-**	**2638**	**0**
Key:	**Positive values are cooling loads Negative values are heating loads**			**Positive values are heating loads Negative values are cooling loads**		

Note 1: Surface convection line items show the combined effects of conductive heat gain to the surface and radiative heat gains absorbed at the surface which are then convected to room air.

Note 2: Lighting, equipment, and people line items include only the direct convective heat gain from the heat source to the room air. The radiative portion of the heat gain is first absorbed by surfaces in the room and then later convected from the surface to the air. Therefore the effect of the radiative portion of the heat gain is found in the surface convection line items.

Note 3: Solar heat gain is absorbed by surfaces in the room, re-radiated to other surfaces, and finally convected from the surfaces to room air. Therefore, the effect of solar heat gain is found in the surface convection line items.

DOAS Sizing Summary for UNIT QQ
(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton) - Copy(converted) 06/11/2026
Prepared by: Shakespeare Engineering 11:07 AM

Air System Information

Air System Name	**UNIT QQ**		Number of zones	**1**	
Equipment Class	**TERM**		Floor Area	**454.7**	sqft
Air System Type	**PKG-FC**		Location	**St George Municipal, UT, USA**	

Sizing Calculation Information

Calculation Months	**Jan to Dec**	Zone CFM Sizing	**Sum of space airflow rates**
Sizing Data	**Calculated**	Space CFM Sizing	**Individual peak space loads**

Cooling Coil Sizing Data

Total coil load	**0.1**	Tons	Load occurs at	**July 15:00**	
Total coil load	**1.6**	MBH	OA DB / WB	**106.2 / 66.2**	F
Total coil load	**374.2**	CFM/Ton	Entering DB / WB	**106.1 / 66.2**	F
Sensible coil load	**1.6**	MBH	Leaving DB / WB	**73.5 / 55.4**	F
Coil CFM at July 15:00	**50**	CFM			
Max coil CFM	**50**	CFM			
Sensible heat ratio	**1.000**				
Water flow @ 10.0 F rise	**N/A**				

Heating Coil Sizing Data

Max coil load	**2.4**	MBH	Load occurs at	**Design Heating**	
Coil CFM at Design Heating	**50**	CFM	Ent. DB / Lvg DB	**20.0 / 68.5**	F
Max coil CFM	**50**	CFM			
Water flow @ 20.0 F drop	**N/A**				

Ventilation Fan Sizing Data

Design CFM	**50**	CFM	Fan motor BHP	**0.03**	BHP
Design CFM/sqft	**0.11**	CFM/sqft	Fan motor kW	**0.02**	kW
			Fan total static	**2.00**	in wg

Outdoor Ventilation Air Data

Design airflow CFM	**50**	CFM	CFM/person	**11.00**	CFM/person
CFM/sqft	**0.11**	CFM/sqft			

Zone Sizing Summary for UNIT QQ
(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton) - Copy(converted) 06/11/2026
Prepared by: Shakespeare Engineering 11:07 AM

Air System Information

Air System Name	**UNIT QQ**	Number of zones	**1**
Equipment Class	**TERM**	Floor Area	**454.7** sqft
Air System Type	**PKG-FC**	Location	**St George Municipal, UT, USA**

Sizing Calculation Information

Calculation Months	**Jan to Dec**	Zone CFM Sizing	**Sum of space airflow rates**
Sizing Data	**Calculated**	Space CFM Sizing	**Individual peak space loads**

Terminal Unit Sizing Data - Cooling

Zone Name	Total Coil Load (MBH)	Sens Coil Load (MBH)	Coil Entering DB / WB (F)	Coil Leaving DB / WB (F)	Water Flow @ 10.0 F (gpm)	Time of Peak Coil Load	Zone CFM/sqft
UNIT QQ	8.3	8.0	73.4 / 56.6	57.4 / 50.2	-	August 14:00	1.11

Terminal Unit Sizing Data - Heating, Fan, Ventilation

Zone Name	Heating Coil Load (MBH)	Heating Coil Ent/Lvg DB (F)	Htg Coil Water Flow @20.0 F (gpm)	Fan Design Airflow (CFM)	Fan Motor (BHP)	Fan Motor (kW)	OA Vent Design Airflow (CFM)
UNIT QQ	3.8	72.0 / 79.8	-	506	0.037	0.030	50

Zone Peak Sensible Loads

Zone Name	Zone Cooling Sensible (MBH)	Time of Peak Sensible Cooling Load	Zone Heating Load (MBH)	Zone Floor Area (sqft)
UNIT QQ	7.4	September 14:00	3.1	454.7

Space Loads and Airflows

Zone Name / Space Name	Cooling Sensible (MBH)	Time of Peak Sensible Load	Air Flow (CFM)	Heating Load (MBH)	Floor Area (sqft)	Space CFM/sqft
UNIT QQ						
UNIT 516-QQ	7.4	September 14:00	506	3.1	454.7	1.11

1. Summary

Ventilation Sizing Method .... **Sum of Space OA Airflows**
Design Ventilation Airflow Rate .... **50** CFM

2. Space Ventilation Analysis

2.1 Zone: UNIT QQ

Zone Name / Space Name	Floor Area (sqft)	Maximum Occupants	Maximum Supply Air (CFM)	Required Outdoor Air (CFM/person)	Required Outdoor Air (CFM/sqft)	Required Outdoor Air (CFM)	Required Outdoor Air (% of supply)	Required Outdoor Air (ACH)	Uncorrected Outdoor Air (CFM)	Direct Exhaust Air (CFM)
UNIT QQ										
UNIT 516-QQ	454.7	4.5	506.1	5.00	0.06	0.0	0.0	0.00	50.0	0.0
Totals	**454.7**	**4.5**	**506.1**						**50.0**	**0.0**

Table 1. System Loads

	DESIGN COOLING - AUGUST 14:00			DESIGN HEATING		
	OA DB / WB 104.9 F / 65.8 F			OA DB / WB 20.2 F / 16.3 F		
COMPONENT LOADS	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**
Zone Conditioning	-	7782	511	-	3843	0
Plenum Load	-	0	0	-	0	0
Exhaust Fan Load	50 CFM	0	-	50 CFM	0	-
Ventilation Load	50 CFM	1552	-189	50 CFM	2514	0
Ventilation Fan Load	50 CFM	74	-	50 CFM	-74	-
Zone Fan Coil Fans Load	-	101	-	-	-101	-
>> Total System Loads	**-**	**9509**	**322**	**-**	**6181**	**0**
Cooling Coil	-	1549	0	-	0	0
Heating Coil	-	0	-	-	2353	-
Terminal Unit Cooling	-	7979	313	-	0	0
Terminal Unit Heating	-	0	-	-	3839	-
>> Total Conditioning	**-**	**9528**	**313**	**-**	**6192**	**0**
Key:	**Positive values are cooling loads Negative values are heating loads**			**Positive values are heating loads Negative values are cooling loads**		

Table 2. Zone Heat Balance Loads

	DESIGN COOLING - AUGUST 14:00			DESIGN HEATING		
	OA DB / WB 104.9 F / 65.8 F			OA DB / WB 20.2 F / 16.3 F		
Zone Heat Balance Component	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**
Exterior Wall Convection	351 sqft	1524	-	351 sqft	917	-
Roof Convection	0 sqft	0	-	0 sqft	0	-
Window Convection	36 sqft	545	-	36 sqft	313	-
Skylight Convection	0 sqft	0	-	0 sqft	0	-
Door Convection	0 sqft	0	-	0 sqft	0	-
Floor Convection	455 sqft	942	-	455 sqft	418	-
Interior Wall Convection	387 sqft	614	-	387 sqft	252	-
Ceiling Convection	455 sqft	1883	-	455 sqft	1156	-
Overhead Lighting Convection	126 W	191	-	0 W	0	-
Task Lighting Convection	0 W	0	-	0 W	0	-
Electric Equipment Convection	455 W	1164	-	0 W	0	-
People Convection	5	314	546	0	0	0
Infiltration	0 CFM	0	0	0 CFM	0	0
Miscellaneous Equipment	-	0	0	-	0	0
Air Internal Energy Change	-	0		-	0	0
Safety Factor	0% / 0%	0	0	0%	0	0
>> Total Zone Loads	**-**	**7177**	**546**	**-**	**3056**	**0**
Key:	**Positive values are cooling loads Negative values are heating loads**			**Positive values are heating loads Negative values are cooling loads**		

Note 1: Surface convection line items show the combined effects of conductive heat gain to the surface and radiative heat gains absorbed at the surface which are then convected to room air.

Note 2: Lighting, equipment, and people line items include only the direct convective heat gain from the heat source to the room air. The radiative portion of the heat gain is first absorbed by surfaces in the room and then later convected from the surface to the air. Therefore the effect of the radiative portion of the heat gain is found in the surface convection line items.

Note 3: Solar heat gain is absorbed by surfaces in the room, re-radiated to other surfaces, and finally convected from the surfaces to room air. Therefore, the effect of solar heat gain is found in the surface convection line items.

DOAS Sizing Summary for UNIT QQ XXL

(In Alternative: Default Alternative)

Air System Information

Air System Name	**UNIT QQ XXL**		Number of zones	**1**	
Equipment Class	**TERM**		Floor Area	**444.8**	sqft
Air System Type	**PKG-FC**		Location	**St George Municipal, UT, USA**	

Sizing Calculation Information

Calculation Months	**Jan to Dec**	Zone CFM Sizing	**Sum of space airflow rates**
Sizing Data	**Calculated**	Space CFM Sizing	**Individual peak space loads**

Cooling Coil Sizing Data

Total coil load	**0.1**	Tons	Load occurs at	**July 15:00**	
Total coil load	**1.6**	MBH	OA DB / WB	**106.2 / 66.2**	F
Total coil load	**374.2**	CFM/Ton	Entering DB / WB	**106.1 / 66.2**	F
Sensible coil load	**1.6**	MBH	Leaving DB / WB	**73.5 / 55.4**	F
Coil CFM at July 15:00	**49**	CFM			
Max coil CFM	**49**	CFM			
Sensible heat ratio	**1.000**				
Water flow @ 10.0 F rise	**N/A**				

Heating Coil Sizing Data

Max coil load	**2.3**	MBH	Load occurs at	**Design Heating**	
Coil CFM at Design Heating	**49**	CFM	Ent. DB / Lvg DB	**20.0 / 68.5**	F
Max coil CFM	**49**	CFM			
Water flow @ 20.0 F drop	**N/A**				

Ventilation Fan Sizing Data

Design CFM	**49**	CFM	Fan motor BHP	**0.03**	BHP
Design CFM/sqft	**0.11**	CFM/sqft	Fan motor kW	**0.02**	kW
			Fan total static	**2.00**	in wg

Outdoor Ventilation Air Data

Design airflow CFM	**49**	CFM	CFM/person	**11.00**	CFM/person
CFM/sqft	**0.11**	CFM/sqft			

Zone Sizing Summary for UNIT QQ XXL

(In Alternative: Default Alternative)

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Prepared by: Shakespeare Engineering 11:08 AM

Air System Information

Air System Name ..... **UNIT QQ XXL**
Equipment Class ..... **TERM**
Air System Type ..... **PKG-FC**

Number of zones **1**
Floor Area ..... **444.8** sqft
Location ..... **St George Municipal, UT, USA**

Sizing Calculation Information

Calculation Months ..... **Jan to Dec**
Sizing Data ..... **Calculated**

Zone CFM Sizing ..... **Sum of space airflow rates**
Space CFM Sizing ..... **Individual peak space loads**

Terminal Unit Sizing Data - Cooling

Zone Name	Total Coil Load (MBH)	Sens Coil Load (MBH)	Coil Entering DB / WB (F)	Coil Leaving DB / WB (F)	Water Flow @ 10.0 F (gpm)	Time of Peak Coil Load	Zone CFM/sqft
UNIT QQ XXL	8.1	7.9	73.4 / 55.9	57.1 / 49.3	-	September 14:00	1.11

Terminal Unit Sizing Data - Heating, Fan, Ventilation

Zone Name	Heating Coil Load (MBH)	Heating Coil Ent/Lvg DB (F)	Htg Coil Water Flow @20.0 F (gpm)	Fan Design Airflow (CFM)	Fan Motor (BHP)	Fan Motor (kW)	OA Vent Design Airflow (CFM)
UNIT QQ XXL	3.7	72.0 / 79.7	-	493	0.036	0.029	49

Zone Peak Sensible Loads

Zone Name	Zone Cooling Sensible (MBH)	Time of Peak Sensible Cooling Load	Zone Heating Load (MBH)	Zone Floor Area (sqft)
UNIT QQ XXL	7.2	September 14:00	2.9	444.8

Space Loads and Airflows

Zone Name / Space Name	Cooling Sensible (MBH)	Time of Peak Sensible Load	Air Flow (CFM)	Heating Load (MBH)	Floor Area (sqft)	Space CFM/sqft
UNIT QQ XXL						
UNIT 500 - QQ XXL	7.2	September 14:00	493	2.9	444.8	1.11

1. Summary

Ventilation Sizing Method .... **Sum of Space OA Airflows**
Design Ventilation Airflow Rate .... **49** CFM

2. Space Ventilation Analysis

2.1 Zone: UNIT QQ XXL

Zone Name / Space Name	Floor Area (sqft)	Maximum Occupants	Maximum Supply Air (CFM)	Required Outdoor Air (CFM/person)	Required Outdoor Air (CFM/sqft)	Required Outdoor Air (CFM)	Required Outdoor Air (% of supply)	Required Outdoor Air (ACH)	Uncorrected Outdoor Air (CFM)	Direct Exhaust Air (CFM)
UNIT QQ XXL										
UNIT 500 - QQ XXL	444.8	4.4	492.9	5.00	0.06	0.0	0.0	0.00	48.9	0.0
Totals	**444.8**	**4.4**	**492.9**						**48.9**	**0.0**

Air System Heat Balance Summary for UNIT QQ XXL

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton) - Copy(converted) 06/11/2026
Prepared by: Shakespeare Engineering 11:08 AM

Table 1. System Loads

	DESIGN COOLING - SEPTEMBER 14:00			DESIGN HEATING		
	OA DB / WB 99.7 F / 62.8 F			OA DB / WB 20.2 F / 16.3 F		
COMPONENT LOADS	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**
Zone Conditioning	-	7701	500	-	3677	0
Plenum Load	-	0	0	-	0	0
Exhaust Fan Load	49 CFM	0	-	49 CFM	0	-
Ventilation Load	49 CFM	1271	-318	49 CFM	2459	0
Ventilation Fan Load	49 CFM	73	-	49 CFM	-73	-
Zone Fan Coil Fans Load	-	99	-	-	-99	-
>> Total System Loads	**-**	**9143**	**182**	**-**	**5965**	**0**
Cooling Coil	-	1263	0	-	0	0
Heating Coil	-	0	-	-	2302	-
Terminal Unit Cooling	-	7892	170	-	0	0
Terminal Unit Heating	-	0	-	-	3674	-
>> Total Conditioning	**-**	**9155**	**170**	**-**	**5975**	**0**
Key:	**Positive values are cooling loads Negative values are heating loads**			**Positive values are heating loads Negative values are cooling loads**		

Table 2. Zone Heat Balance Loads

	DESIGN COOLING - SEPTEMBER 14:00			DESIGN HEATING		
	OA DB / WB 99.7 F / 62.8 F			OA DB / WB 20.2 F / 16.3 F		
Zone Heat Balance Component	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**
Exterior Wall Convection	312 sqft	1394	-	312 sqft	815	-
Roof Convection	0 sqft	0	-	0 sqft	0	-
Window Convection	36 sqft	617	-	36 sqft	313	-
Skylight Convection	0 sqft	0	-	0 sqft	0	-
Door Convection	0 sqft	0	-	0 sqft	0	-
Floor Convection	445 sqft	1102	-	445 sqft	408	-
Interior Wall Convection	348 sqft	609	-	348 sqft	226	-
Ceiling Convection	445 sqft	1821	-	445 sqft	1115	-
Overhead Lighting Convection	123 W	187	-	0 W	0	-
Task Lighting Convection	0 W	0	-	0 W	0	-
Electric Equipment Convection	445 W	1138	-	0 W	0	-
People Convection	4	307	534	0	0	0
Infiltration	0 CFM	0	0	0 CFM	0	0
Miscellaneous Equipment	-	0	0	-	0	0
Air Internal Energy Change	-	0		-	0	0
Safety Factor	0% / 0%	0	0	0%	0	0
>> Total Zone Loads	**-**	**7174**	**534**	**-**	**2876**	**0**
Key:	**Positive values are cooling loads Negative values are heating loads**			**Positive values are heating loads Negative values are cooling loads**		

Note 1: Surface convection line items show the combined effects of conductive heat gain to the surface and radiative heat gains absorbed at the surface which are then convected to room air.

Note 2: Lighting, equipment, and people line items include only the direct convective heat gain from the heat source to the room air. The radiative portion of the heat gain is first absorbed by surfaces in the room and then later convected from the surface to the air. Therefore the effect of the radiative portion of the heat gain is found in the surface convection line items.

Note 3: Solar heat gain is absorbed by surfaces in the room, re-radiated to other surfaces, and finally convected from the surfaces to room air. Therefore, the effect of solar heat gain is found in the surface convection line items.